Understand AI Better Than Your Students

Dr. Paul Savala — Associate Professor of Mathematics

Setting the Stage

Common Perceptions of AI

> “AI can think like a human, and will replace humans in all kinds of fields within the next few years.”

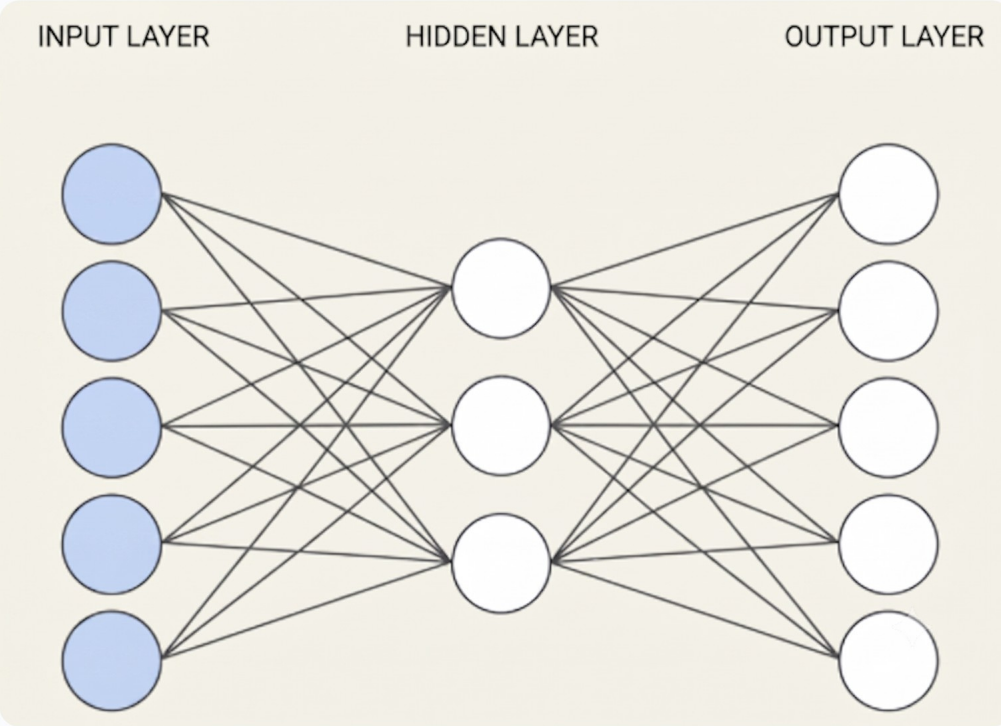
> “AI just repeats back things they found on the internet; they’re basically fancy search engines.”

> “AI constantly makes mistakes, and can’t be relied on for important tasks.”

The Complicated Truth

Understanding the actual systems helps you understand:

- How AI works
- Where it excels
- Where it struggles



Our Focus Today

Not

“How do I use AI?”

But

“How does AI work?”

Questions for Today

1. **What *are* LLMs?**
2. Why are they **good at some things** (coding) but **bad at others** (creative writing)?
3. In what ways **are humans involved**? In what ways **do models "learn on their own"**?

“Human LLM” Activity

Activity Instructions

1. In the top-left corner of your postcard, write one word.
2. Pass it along (snaking through the room).
3. With the postcard you receive, add one more word.
4. Repeat.

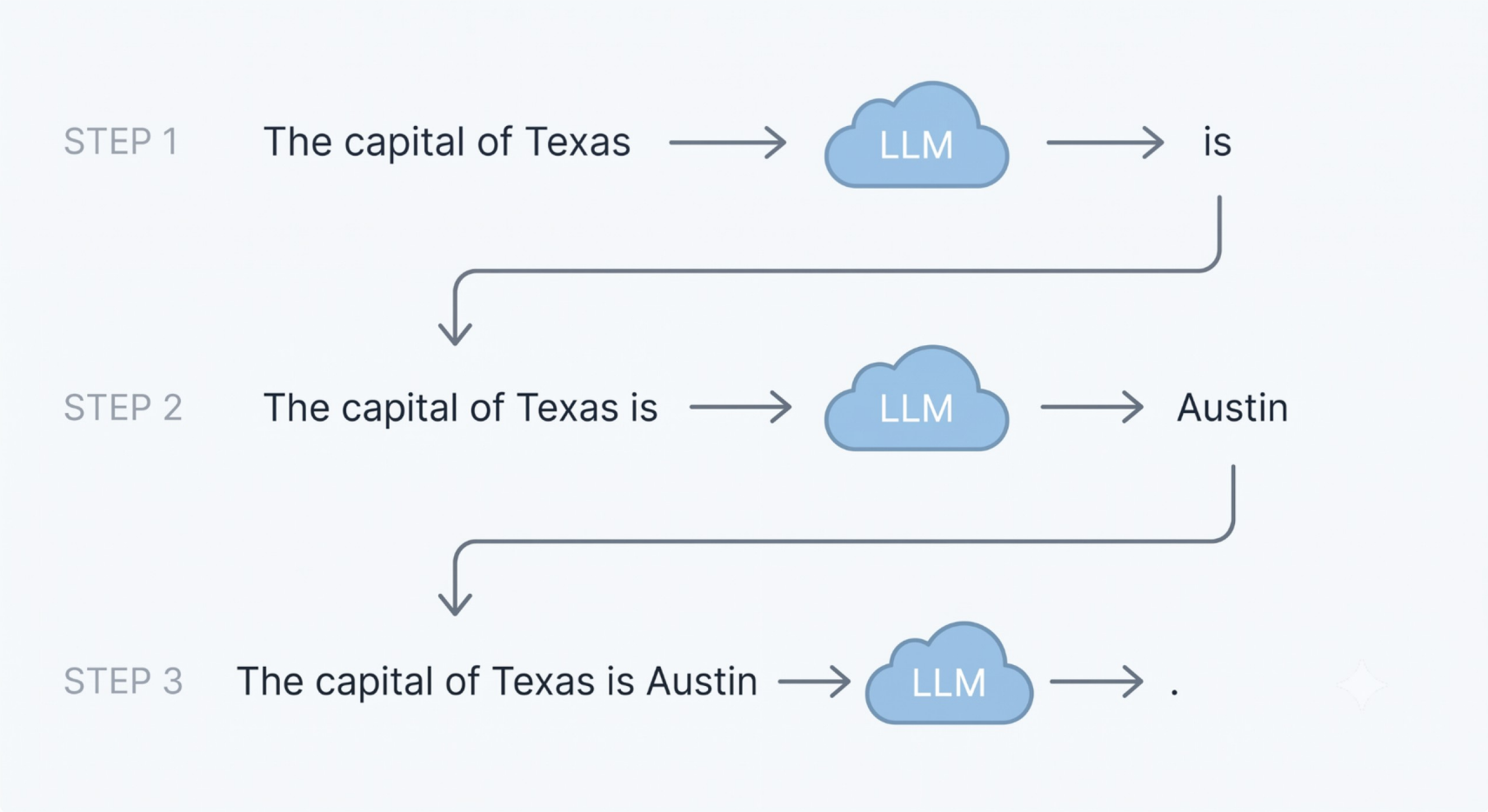
Reflection

What you just did is called **autoregressive generation**.

You generated new words using the previous words as input.

This is how LLMs generate text.

Autoregressive Generation



But what *are* LLMs?

Neural Networks

The Scale of Operations

100 trillion

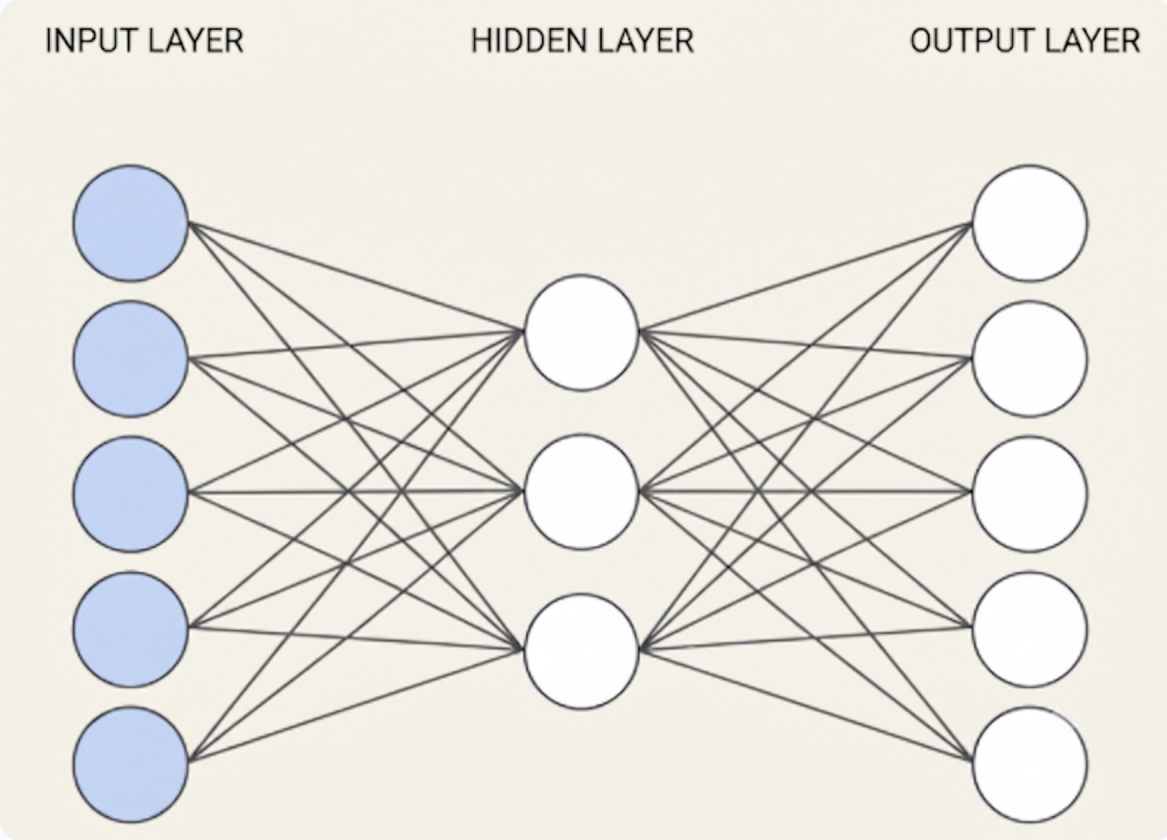
additions and multiplications to answer one question

AI models are **simple math equations**, nothing more.

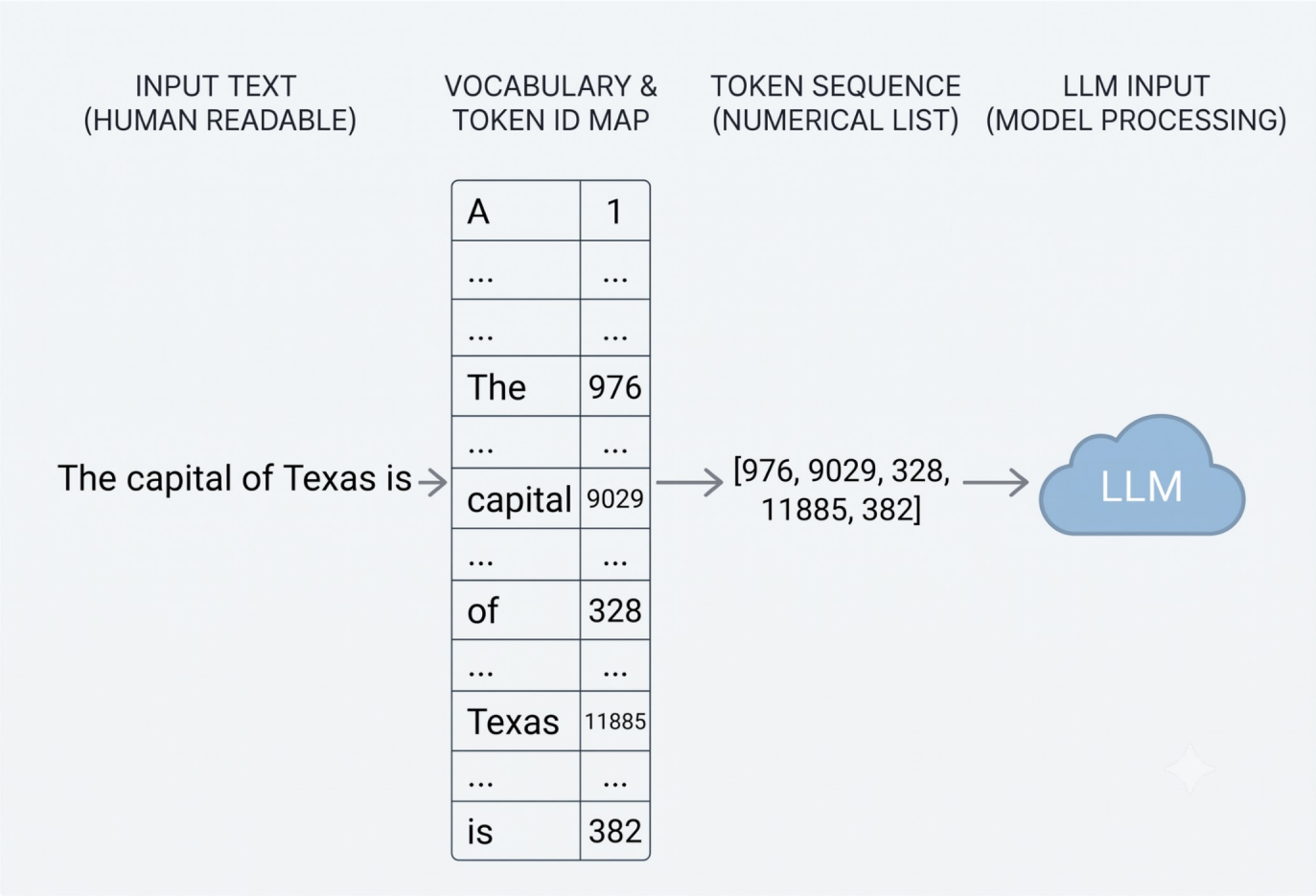
Converting Words to Numbers

A (simplified) **neural network**.

Neural networks only **work with numbers**.



Tokenization



What the Model Actually Sees

Common words

The	capital	of	Texas	is	Austin	.
976	9029	328	11885	382	31628	13

Less common words

M	ito	ch	ond	ria	are	the	powerhouse	of	the	cell	.
44	2955	332	1415	18784	553	290	127147	328	290	4206	13

Try it yourself: platform.openai.com/tokenizer

Trouble Spots: Numbers & Letters

Arithmetic

384	7	×		921	5	=		354	501	05

Spelling

How	many	r	's	are	in	strawberry	?
5299	1991	428	885	553	306	101830	30

Trouble Spots: Scientific Text

DNA

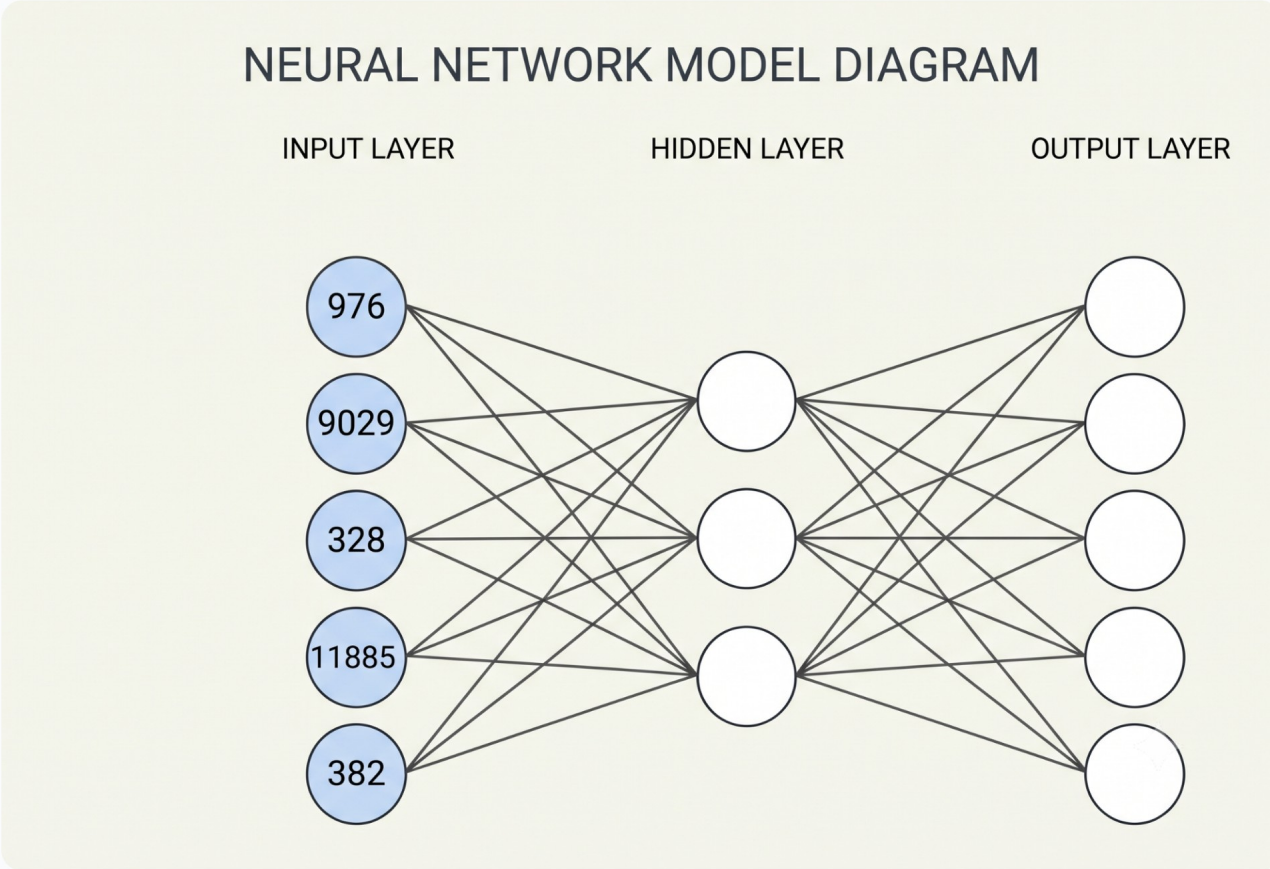
Tokens	AT G CG T AC G TT AG CAT G CA
Codons	ATG CGT ACG TTA GCA TGC A

Chemistry (aspirin)

CC (= O) OC 1 = CC = CC =C 1 C (= O) O

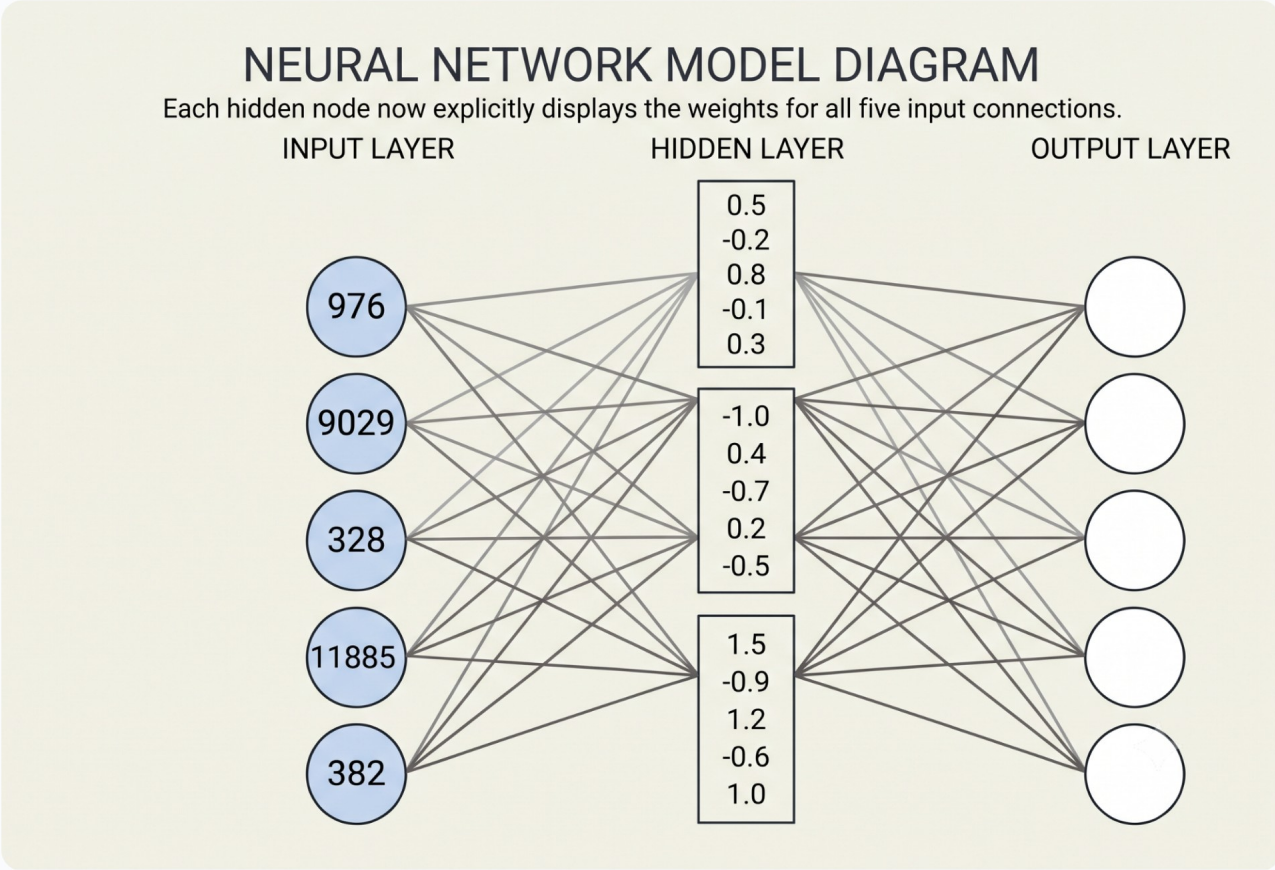
Inside the Network

The token IDs go into the **input nodes**.



Inside the Network

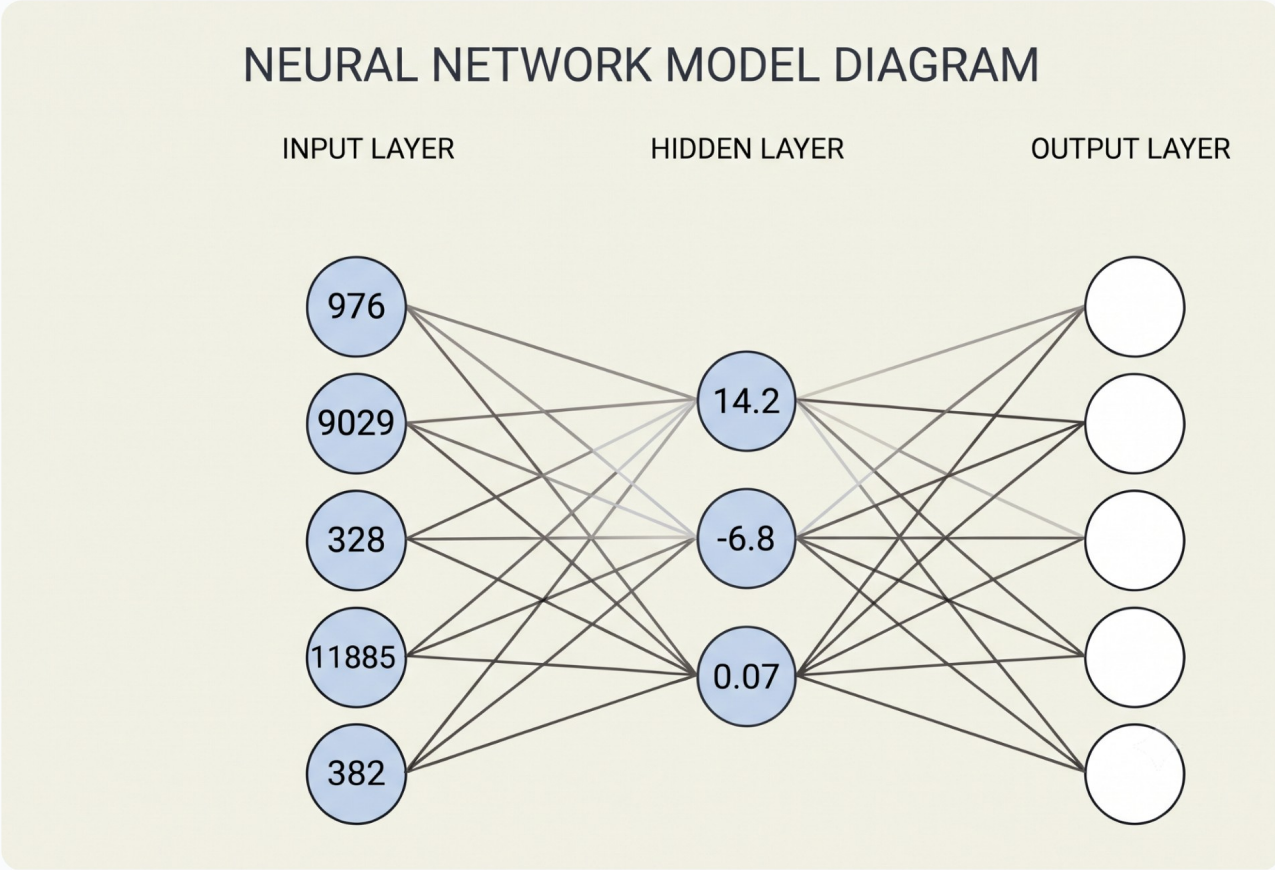
Each hidden node comes with numbers called **parameters** (weights).



Inside the Network

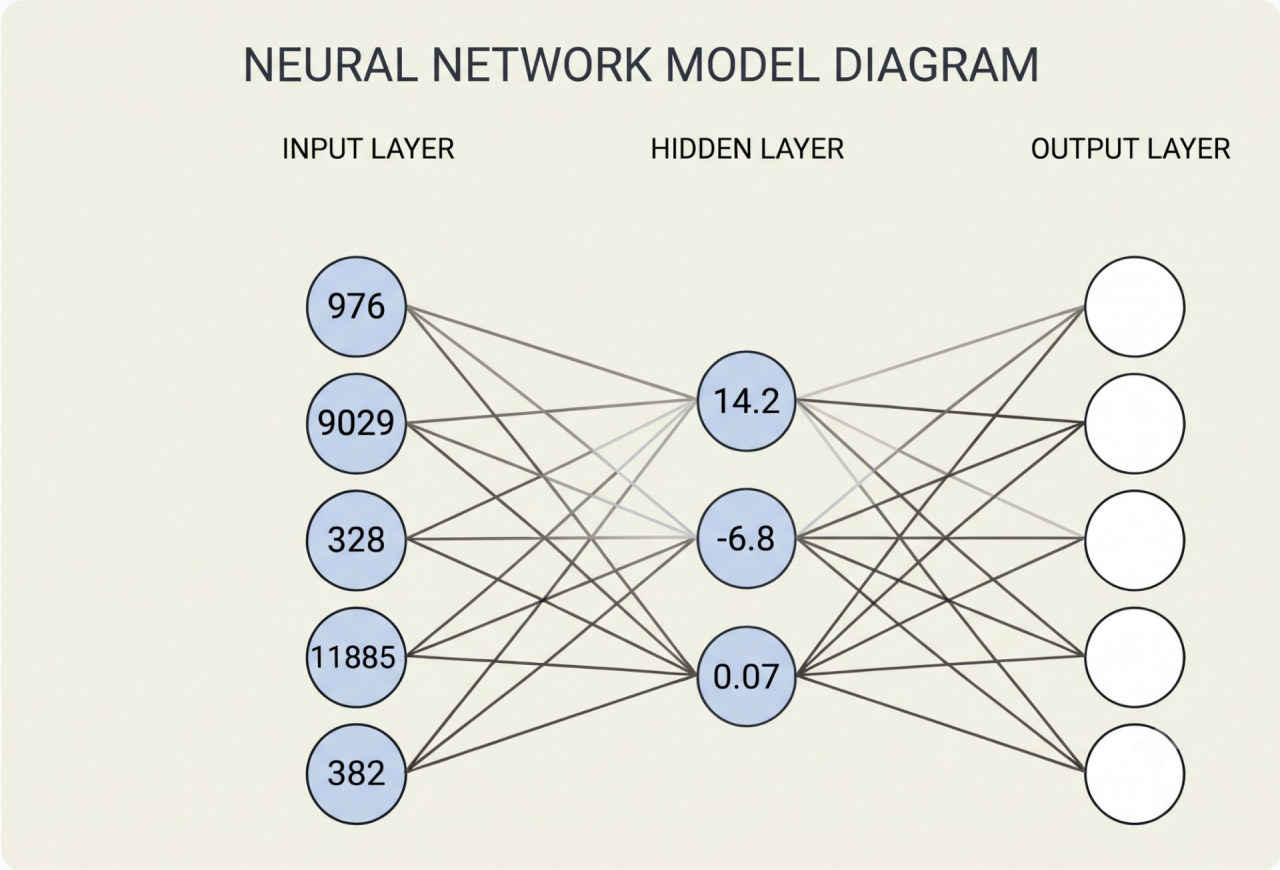
Multiply each input by its weight, then add.

```
  976 × 0.5
+ 9029 × (-0.2)
+ 328 × 0.8
+ …
= 14.2
```



Inside the Network

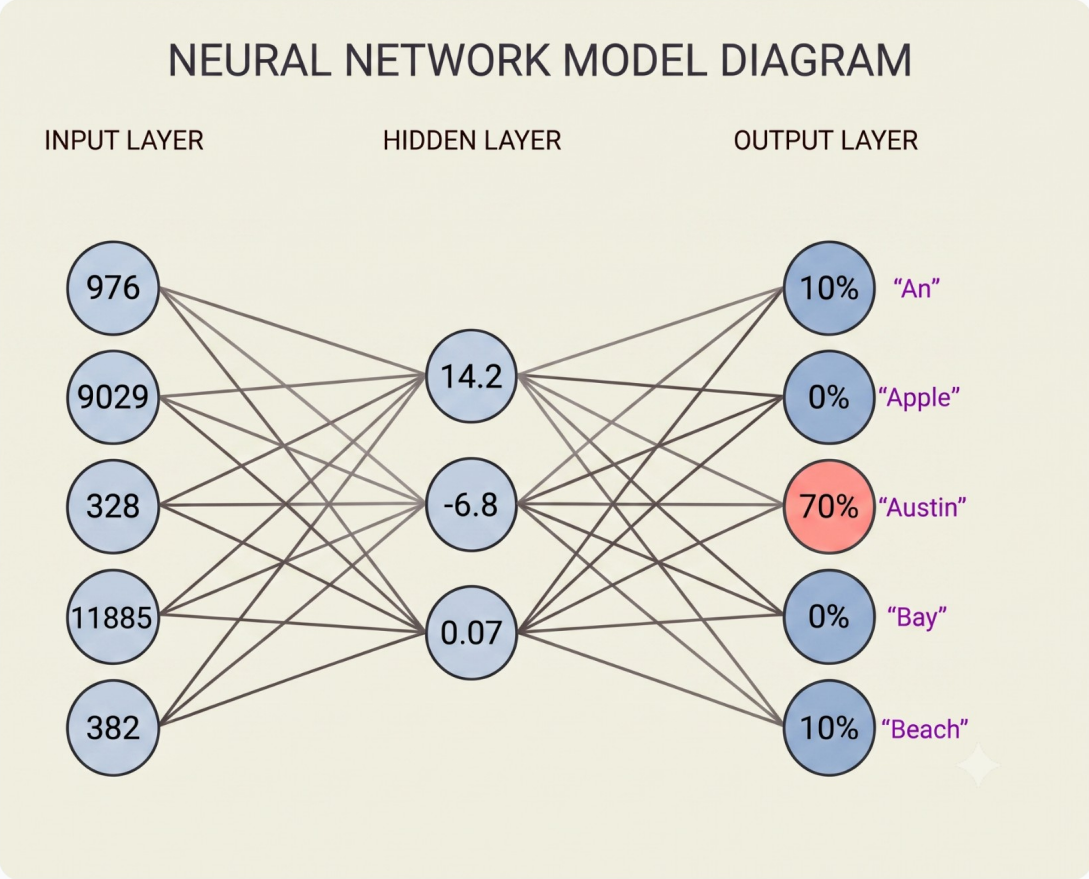
Repeat for every node, layer after layer.



Inside the Network

The outputs are the model's **guesses for the next word**.

Most likely: **"Austin"**



Key Takeaways

Numbers & Math

AI models are just numbers. They convert your input into numbers and perform simple mathematical operations on them.

Prediction

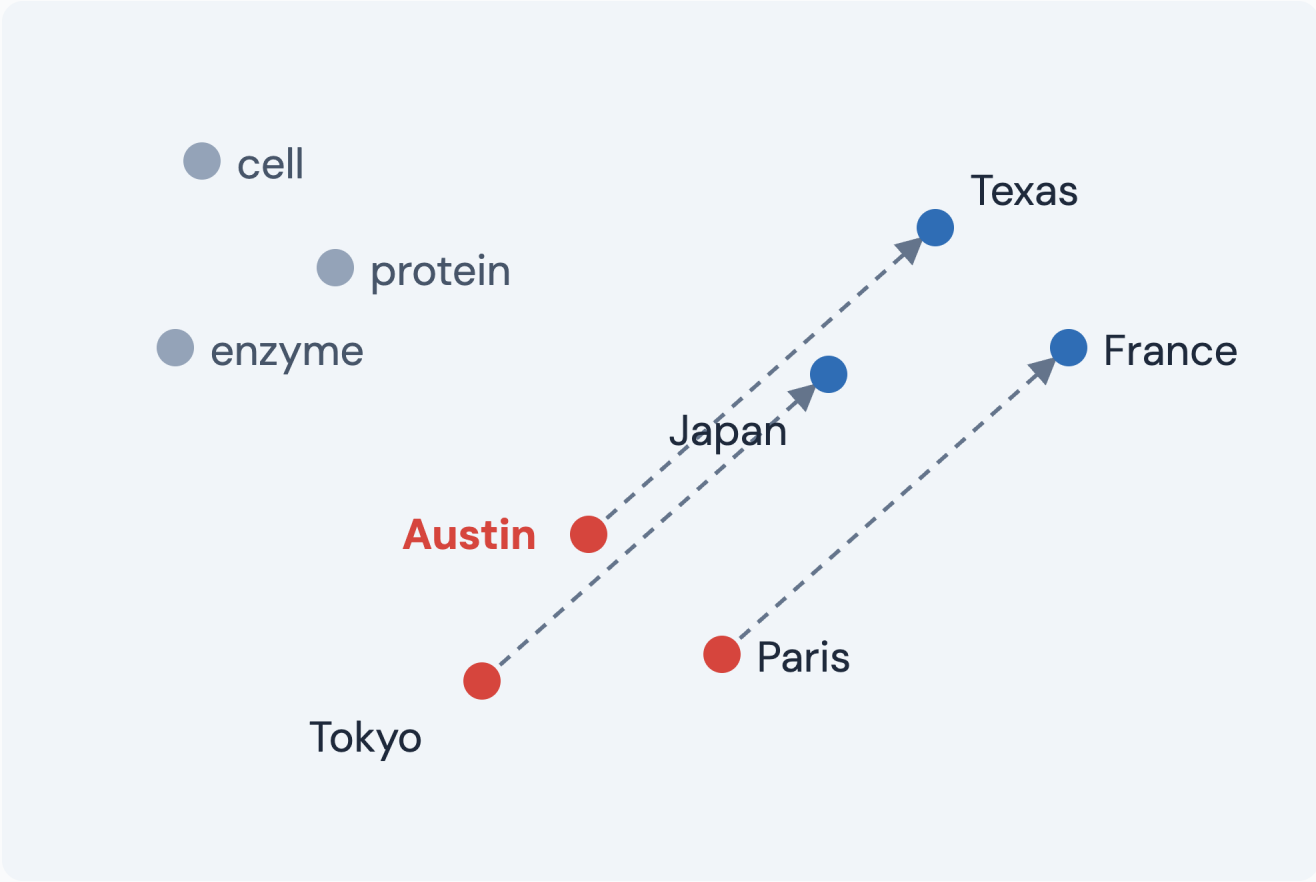
The model predicts which next word is most likely to follow the input words (prompt).

```
Your prompt: “Write me a story about a young boy.”
Model, to itself: What is the most likely first word? Perhaps “Once (upon a time)…”
```

Words as Points in Space

```
Austin →
[0.12, -0.48, 0.91,
 …, 0.07]
```

thousands of numbers per token



Learned by Filling in Blanks

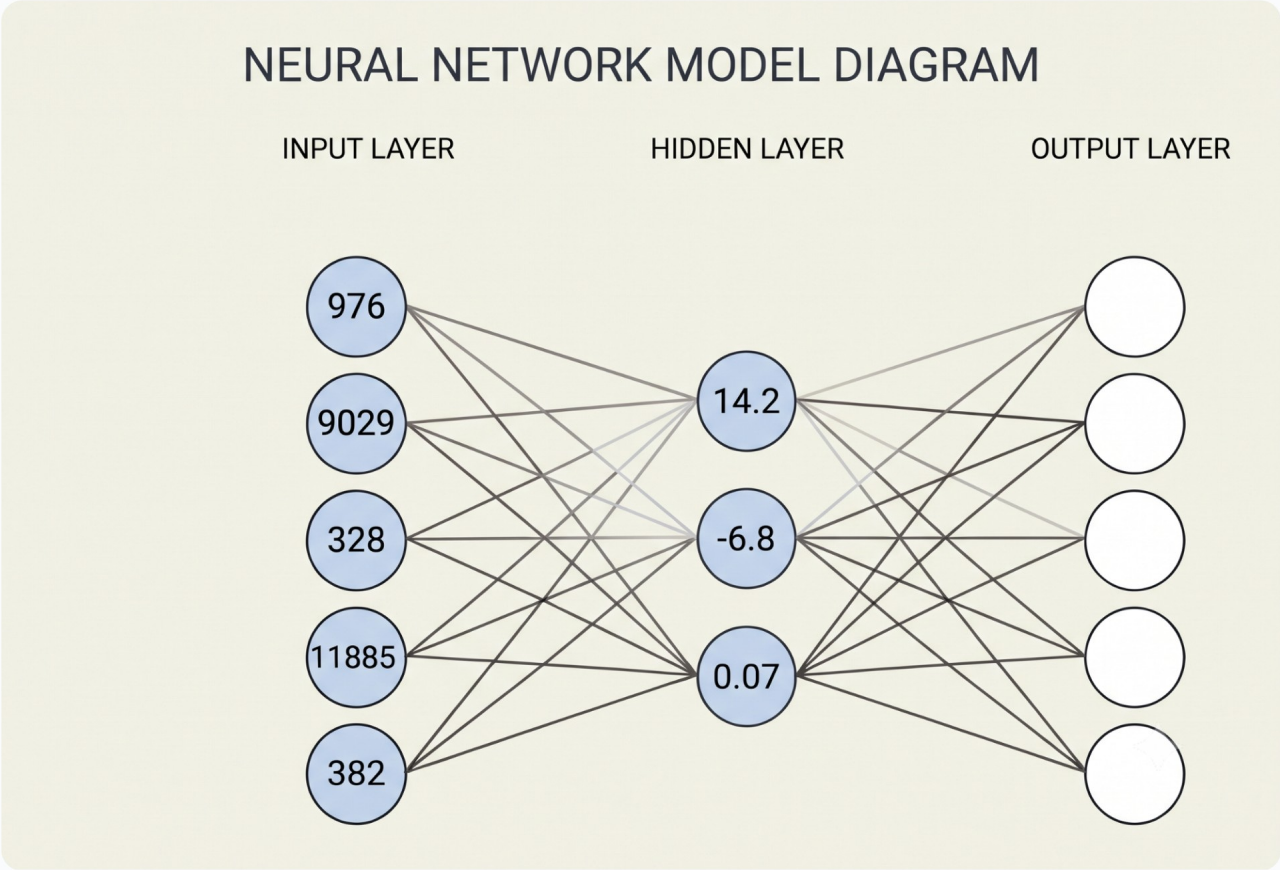
She flew from _____ to Dallas.	Austin, Paris, Tokyo
The _____ folds into its active shape.	protein, enzyme
I added _____ to my coffee.	milk, cream, sugar

Words that fit the same blanks end up **close together**.

Where Do Embeddings Come From?

The numbers in the **hidden layer** are the embedding.

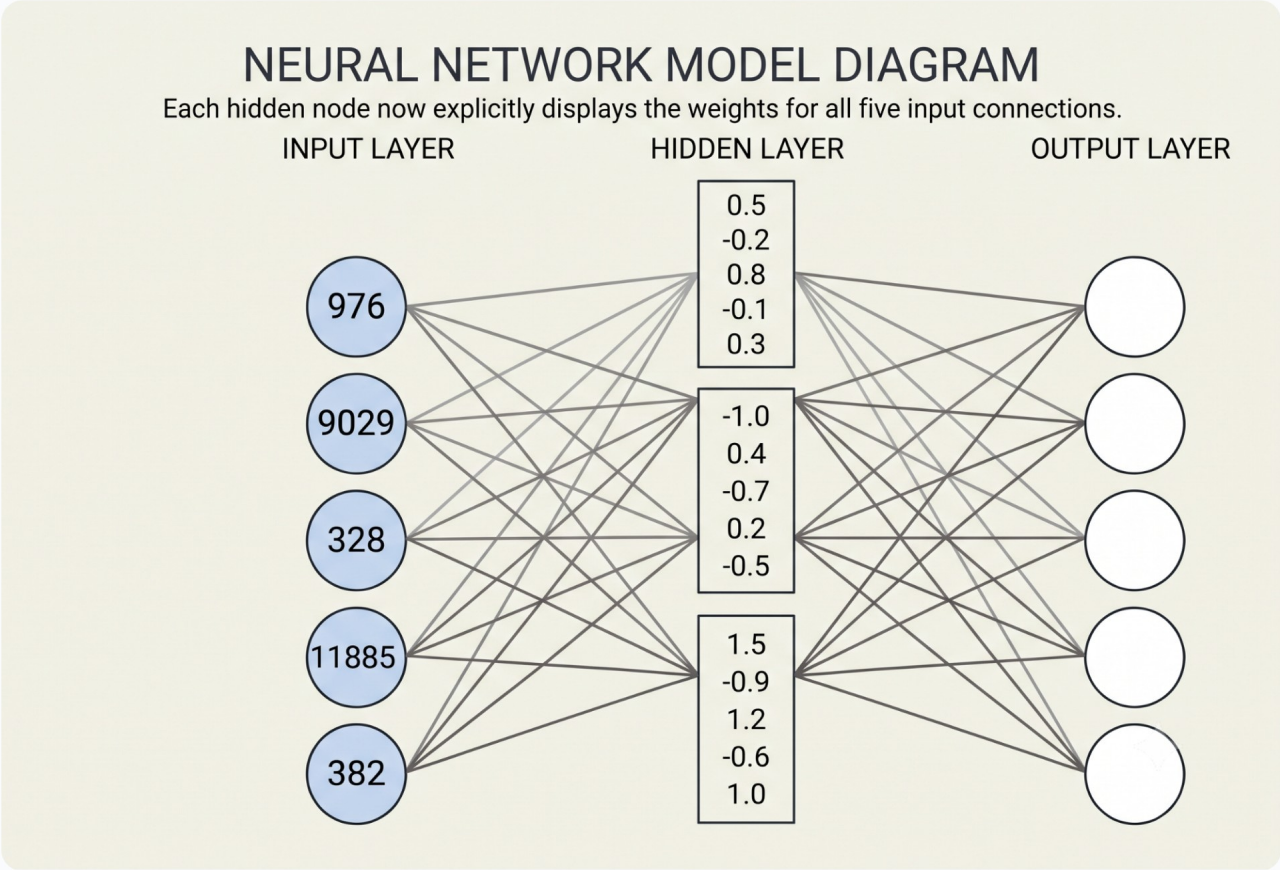
```
[14.2, -6.8, 0.07]
```



But how does the LLM *know* what word should come next?

The Magic Comes from the Weights

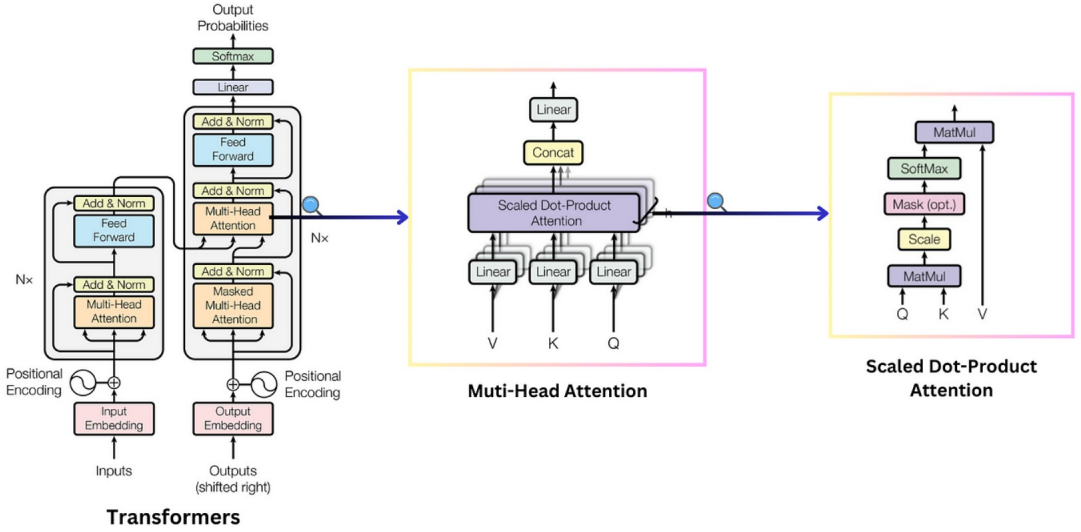
Our simple network: **15 parameters**.



Modern LLMs

1 trillion+

parameters



Picking Good Weights

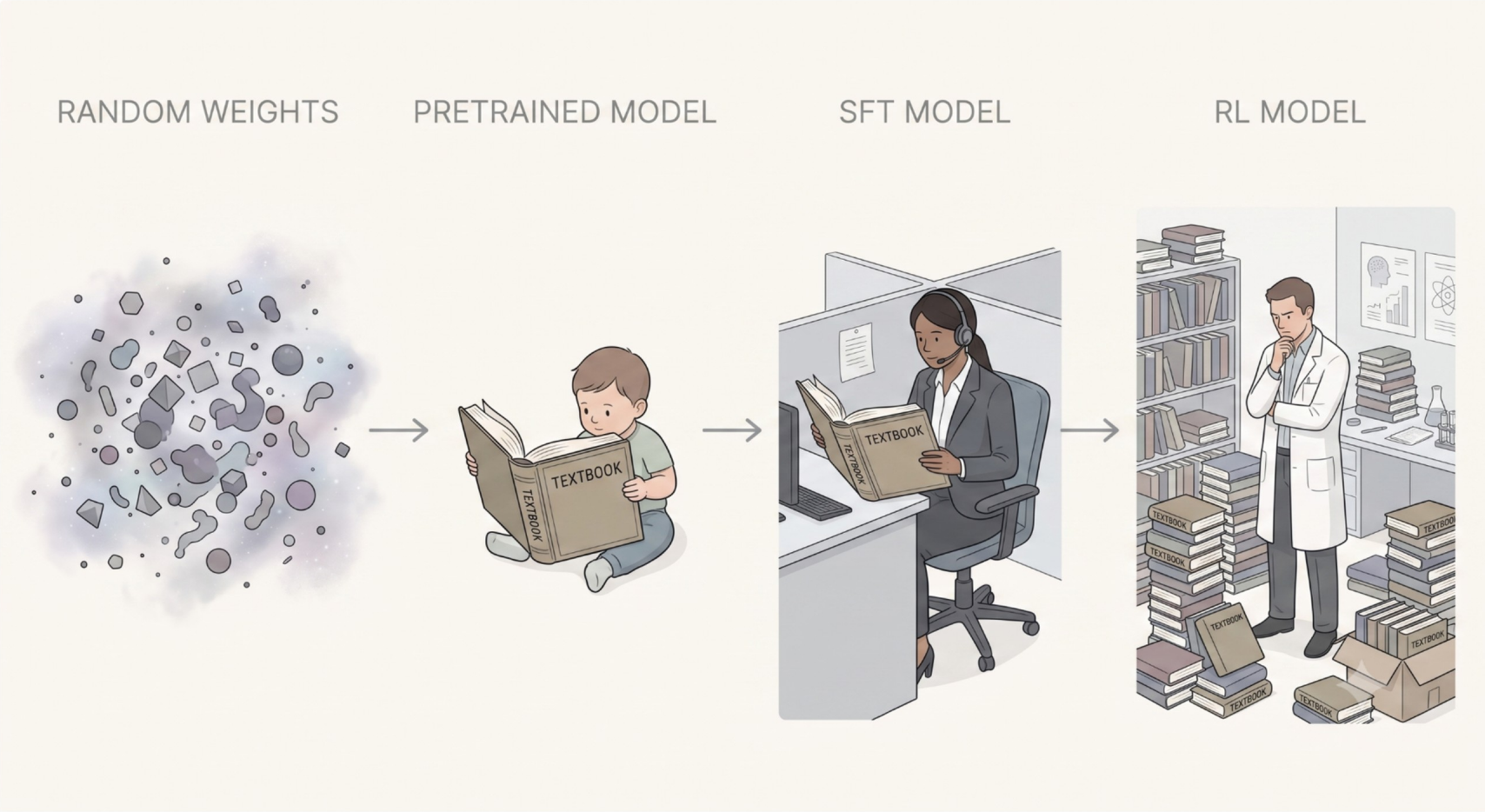
Initially, the weights are just **random numbers**.

We have to **train** the model to pick good weights.

Good weights mean good predictions.

Training Language Models

The Training Process

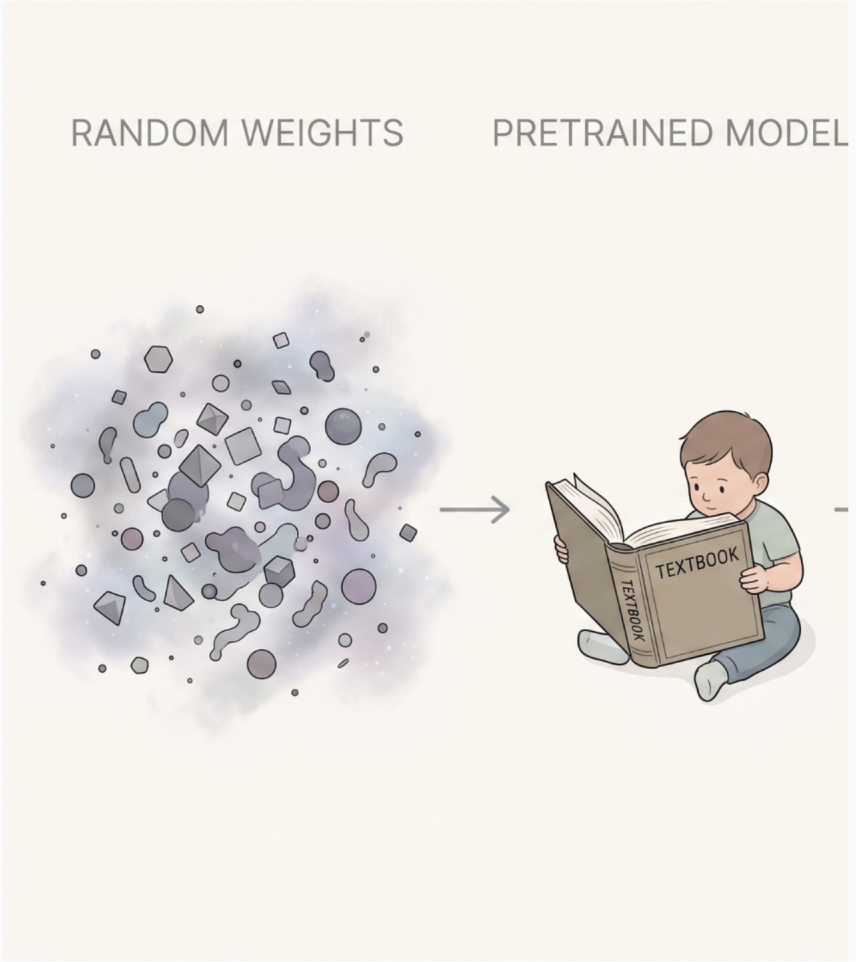


1. Pretraining

15–20 trillion

words of text used to train GPT-5

Training: see a snippet of text, **predict the next word**.



Step 1: Guess

The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___

Untrained model's top guesses

`keen` `íf` `PN` `much` `wild`

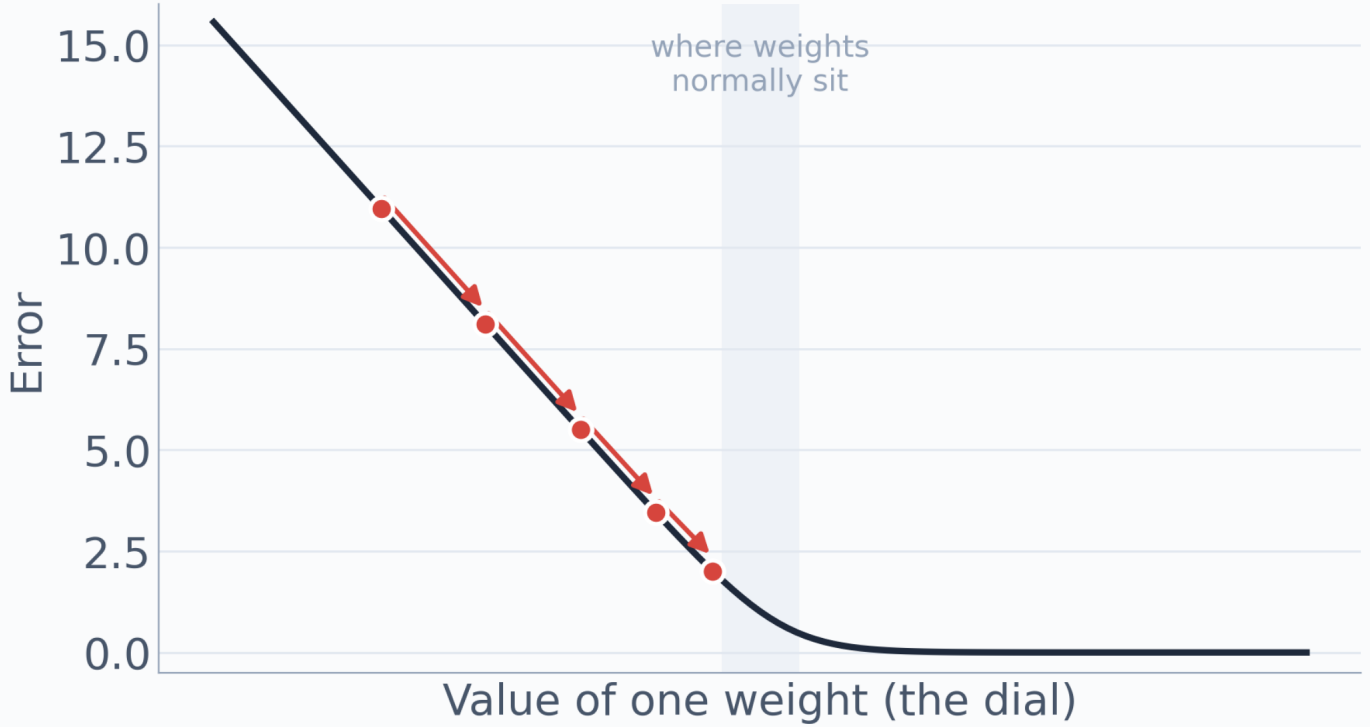
"Austin": **0.0015%**

Step 2: Measure the Error

Checkpoint	P("Austin")	Error
Step 0	**0.0015%**	11.1
Step 2,000	**0.04%**	7.8
Step 8,000	**17%**	1.8
Step 143,000	**36%**	1.0
Perfect	**100%**	0

Step 3: Turn the Dial Downhill

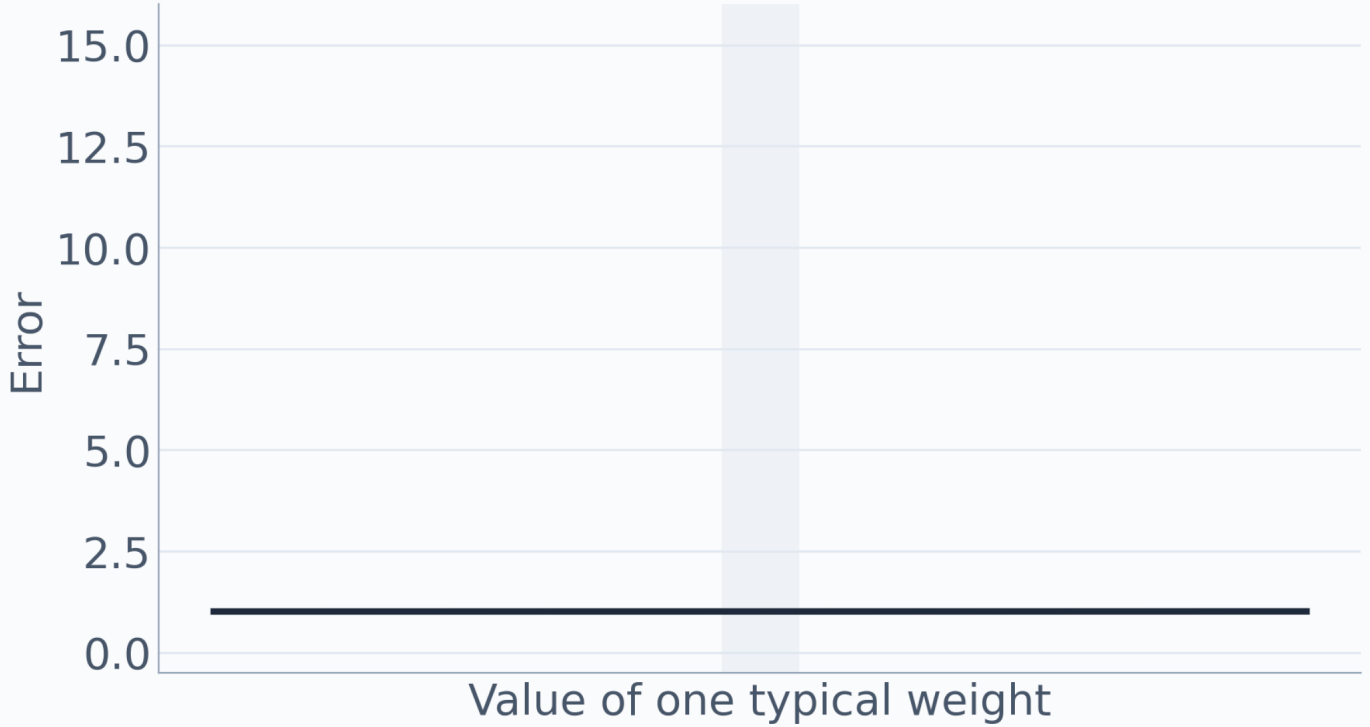
One weight = **one dial**.



But Most Single Dials Barely Matter

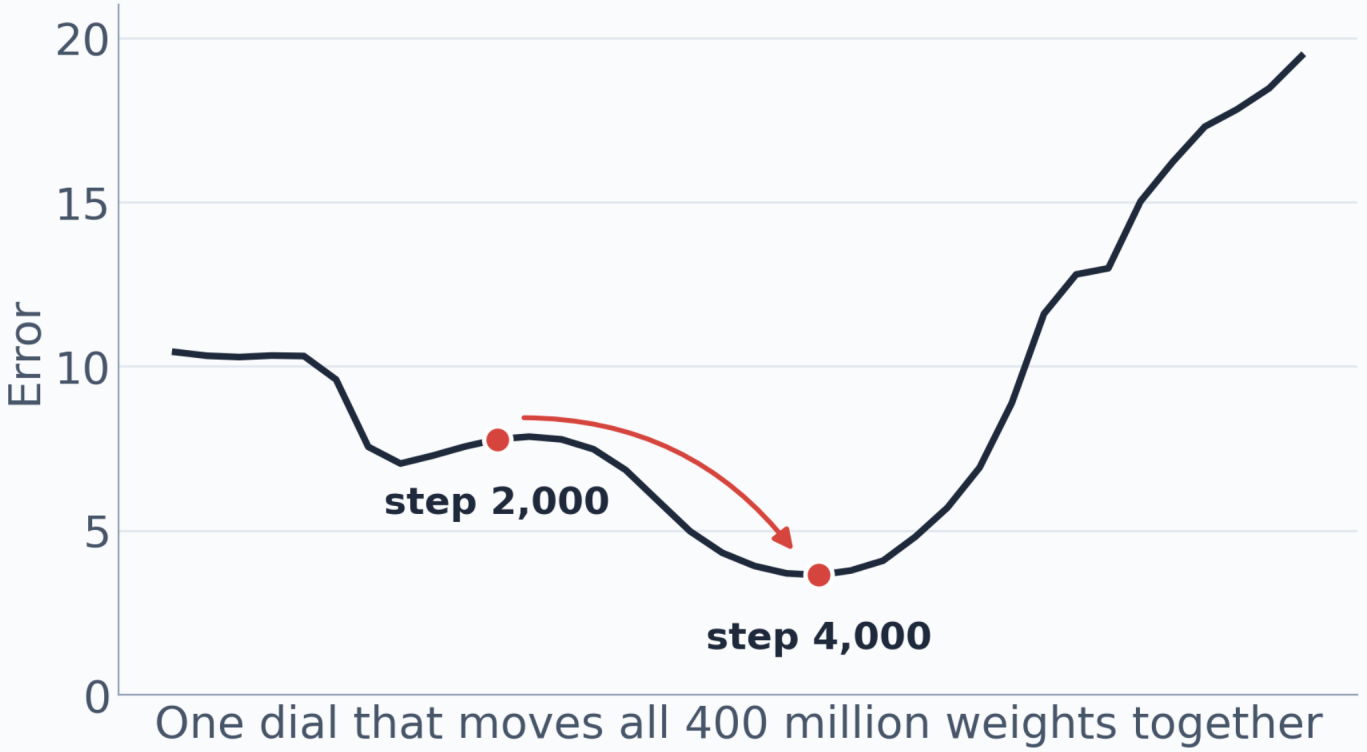
400 million

weights in this model



So Training Turns Every Dial at Once

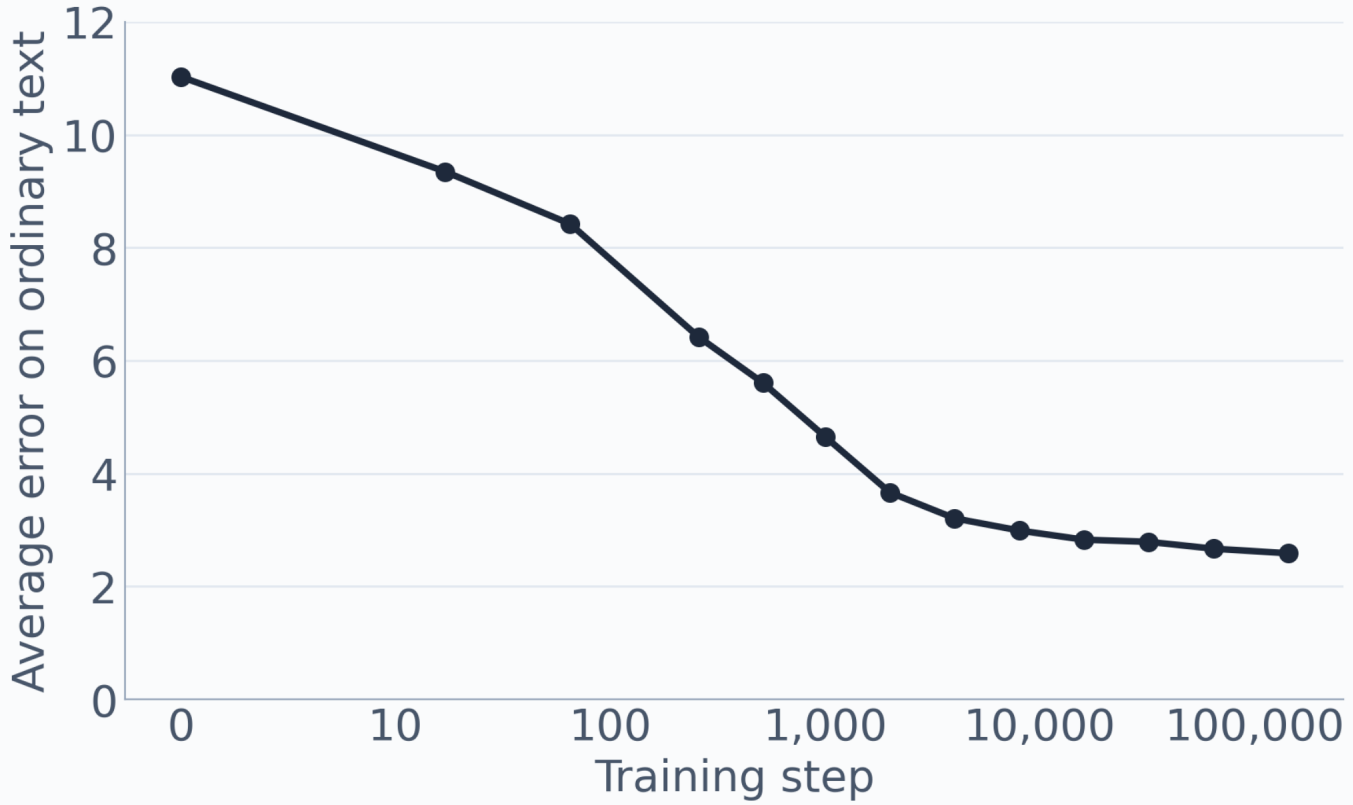
Each step nudges **all** the weights a tiny amount.



Then Repeat, 143,000 Times

143,000
training steps

300 billion
tokens of text

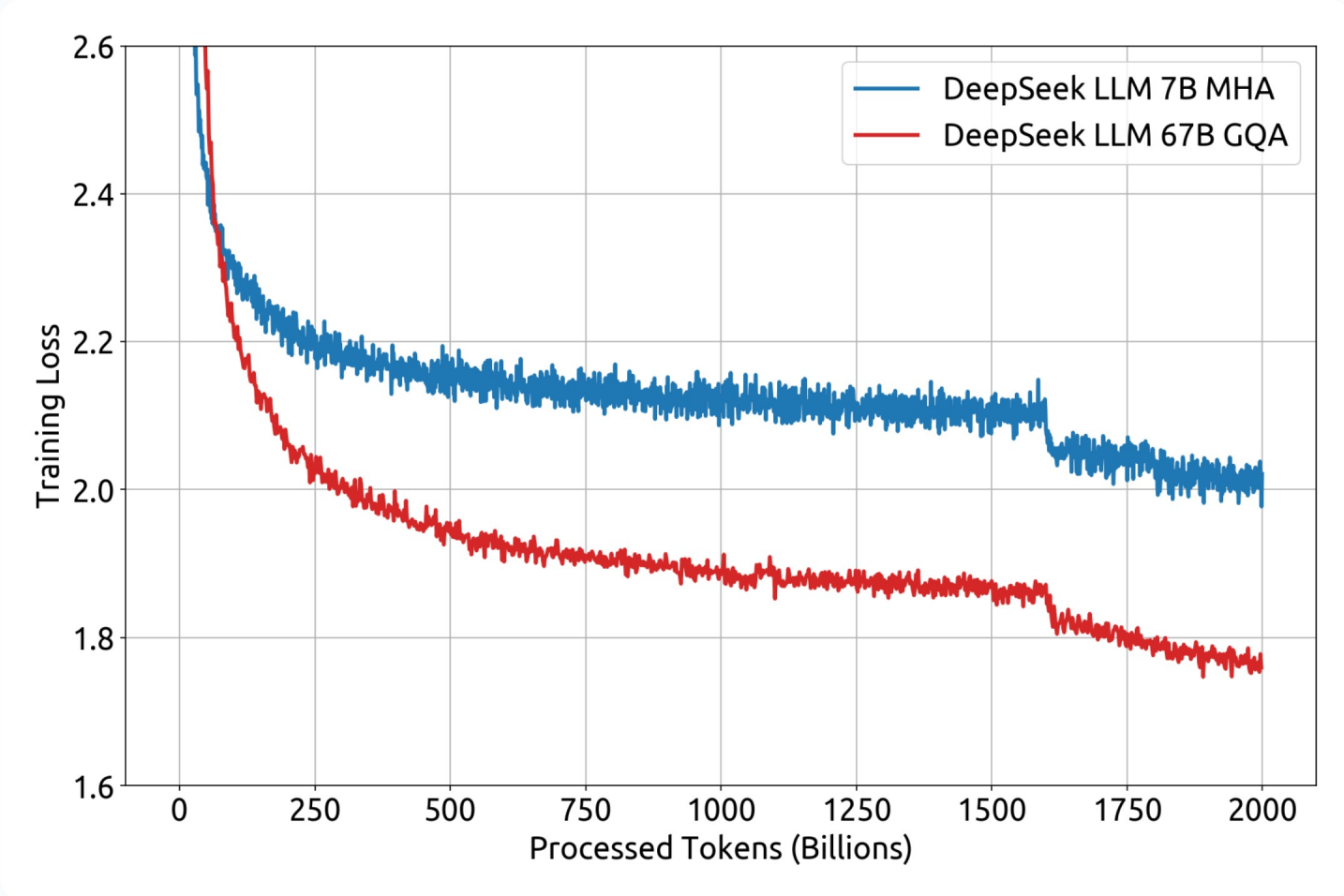


Same Shape at Frontier Scale

2 trillion

tokens of text

DeepSeek LLM 7B and 67B (DeepSeek-AI, 2023)



Our Real Model: Pythia-410M

410 million
weights

300 billion
tokens of training text
(trained by EleutherAI)

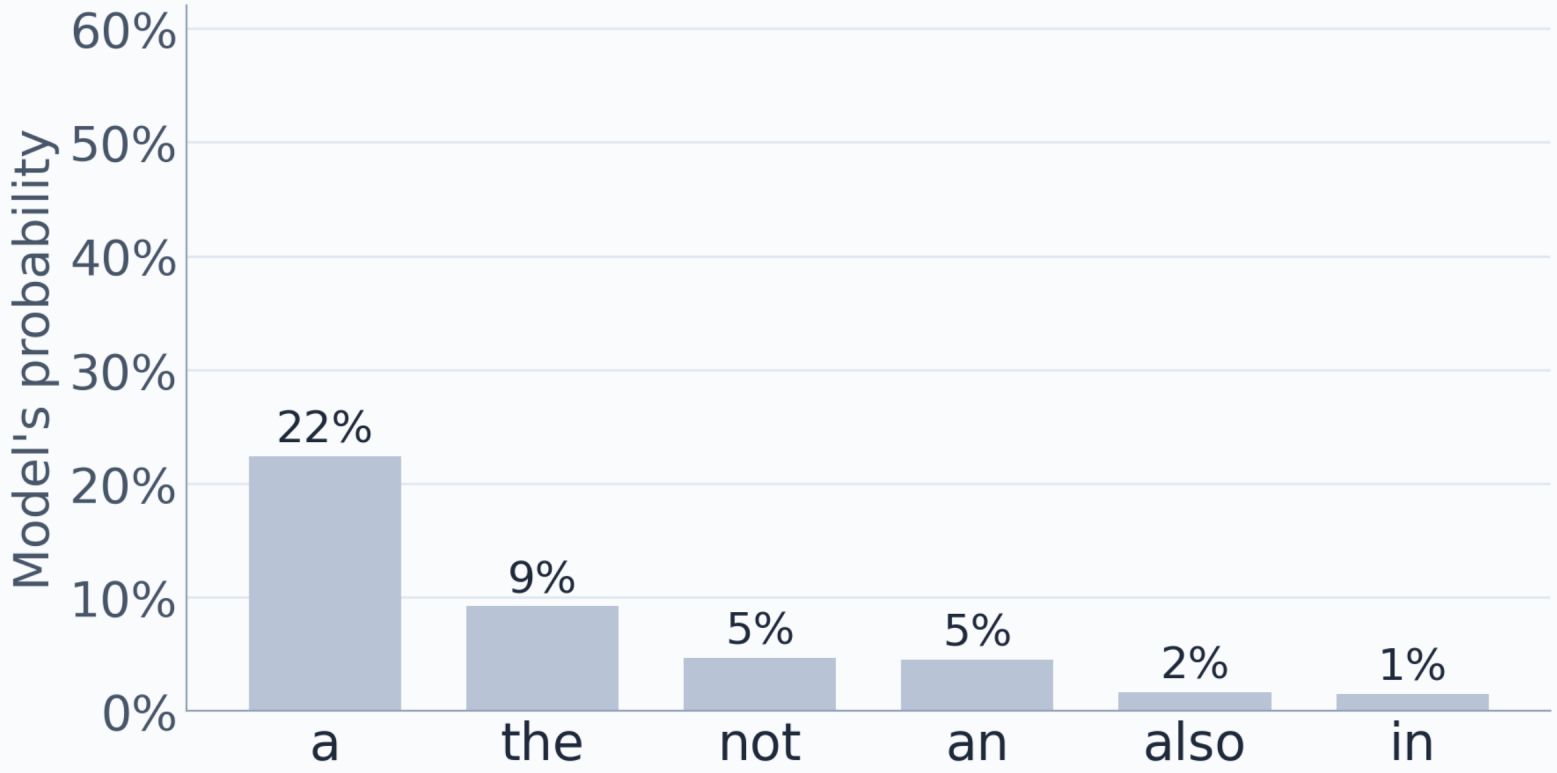
154
snapshots saved
during training

We downloaded 13 snapshots to a laptop and asked each one to finish:

> The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___

Watching a Real Model Learn

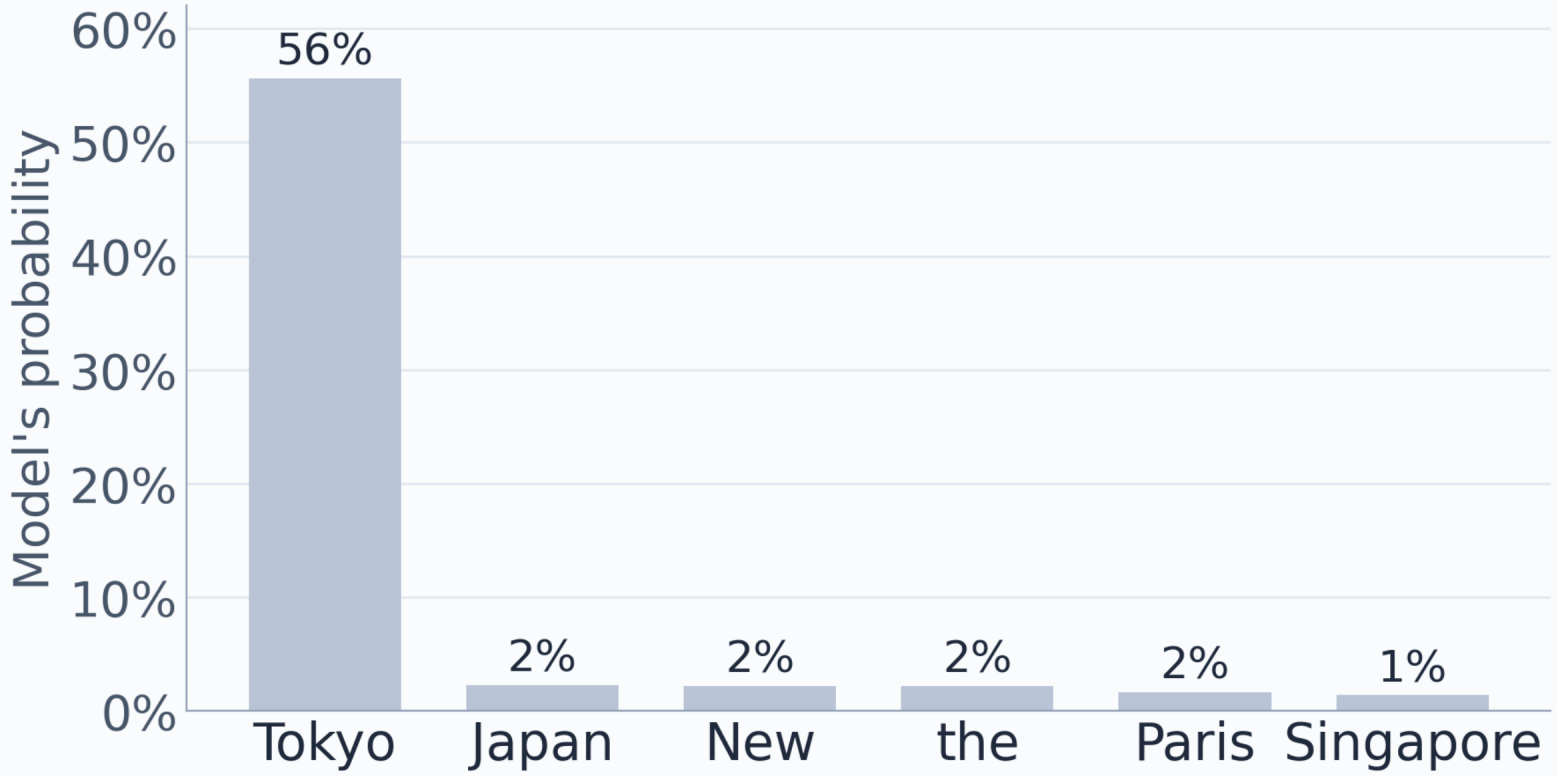
The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___



Step 512

Watching a Real Model Learn

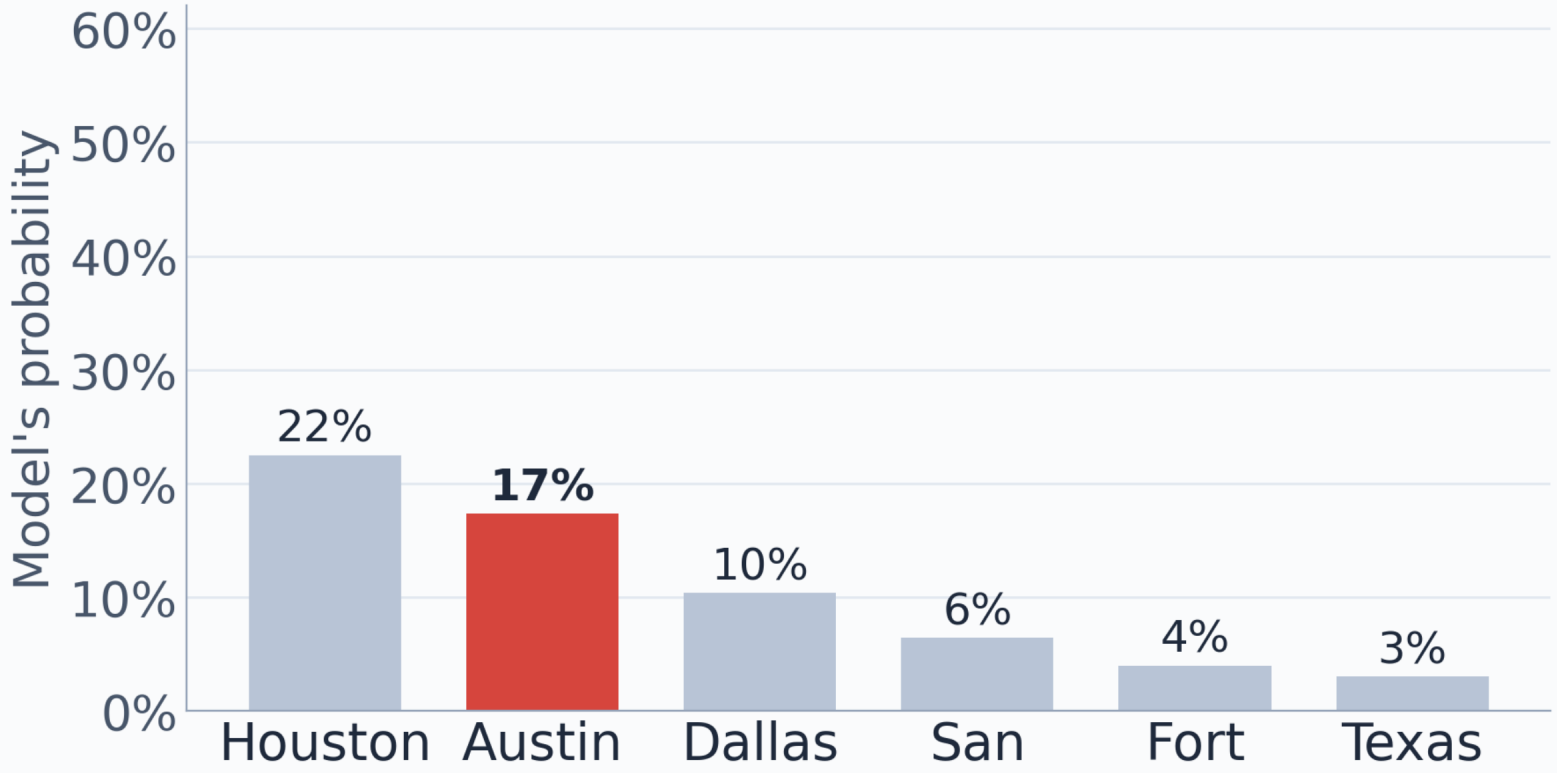
The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___



Step 2,000

Watching a Real Model Learn

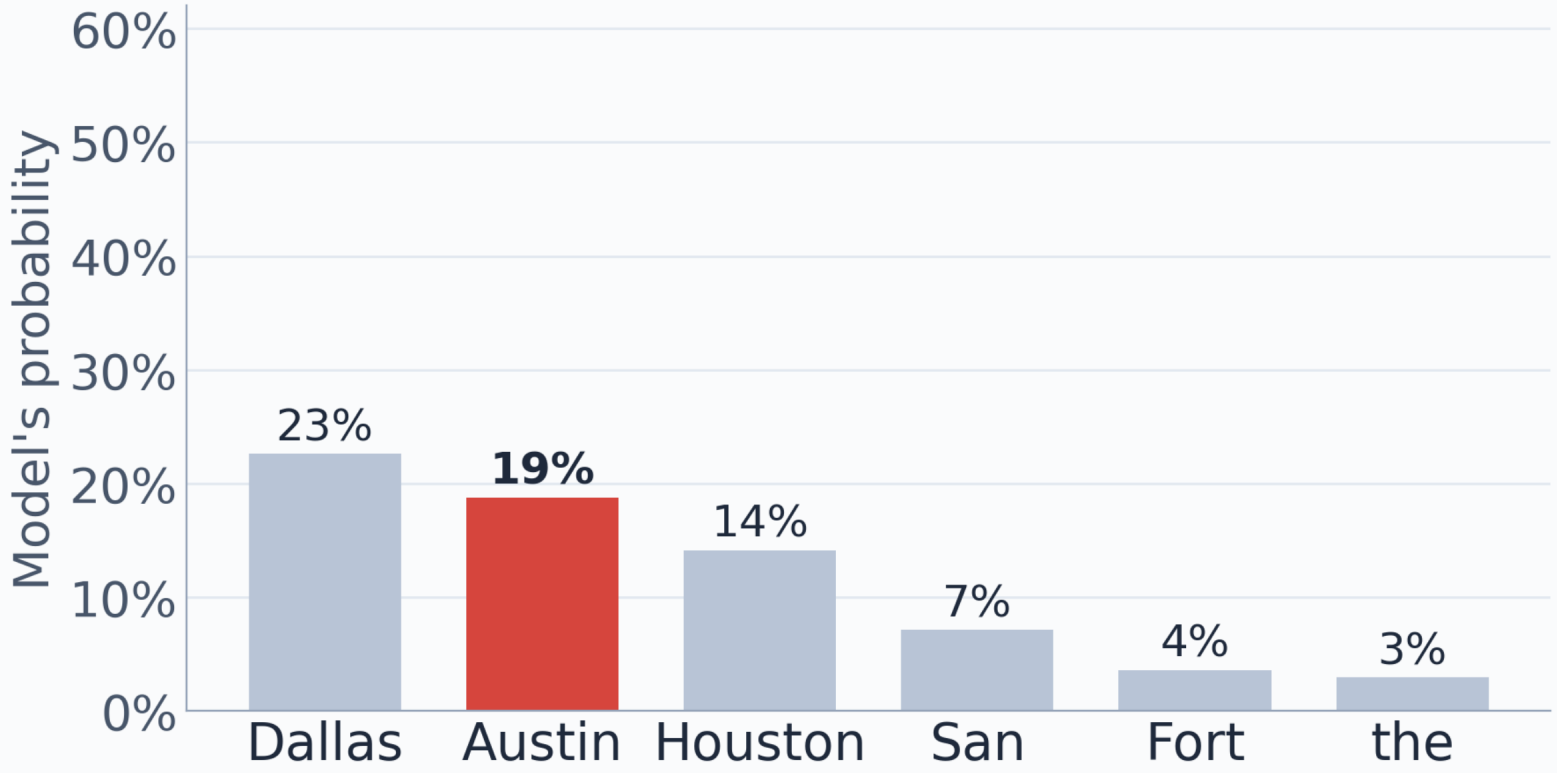
The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___



Step 8,000

Watching a Real Model Learn

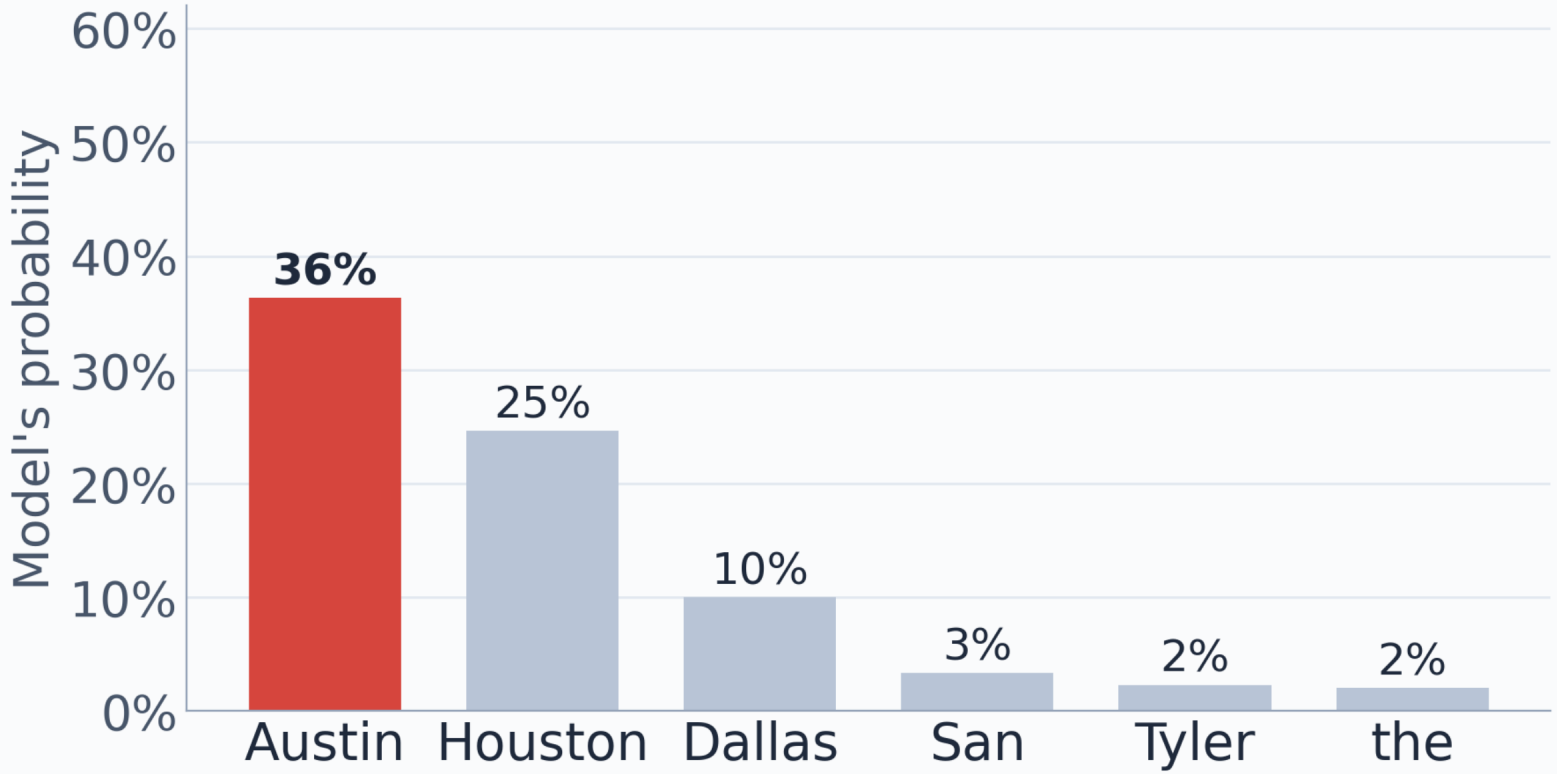
The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___



Step 16,000

Watching a Real Model Learn

The capital of France is Paris. The capital of Japan is Tokyo. The capital of Texas is ___

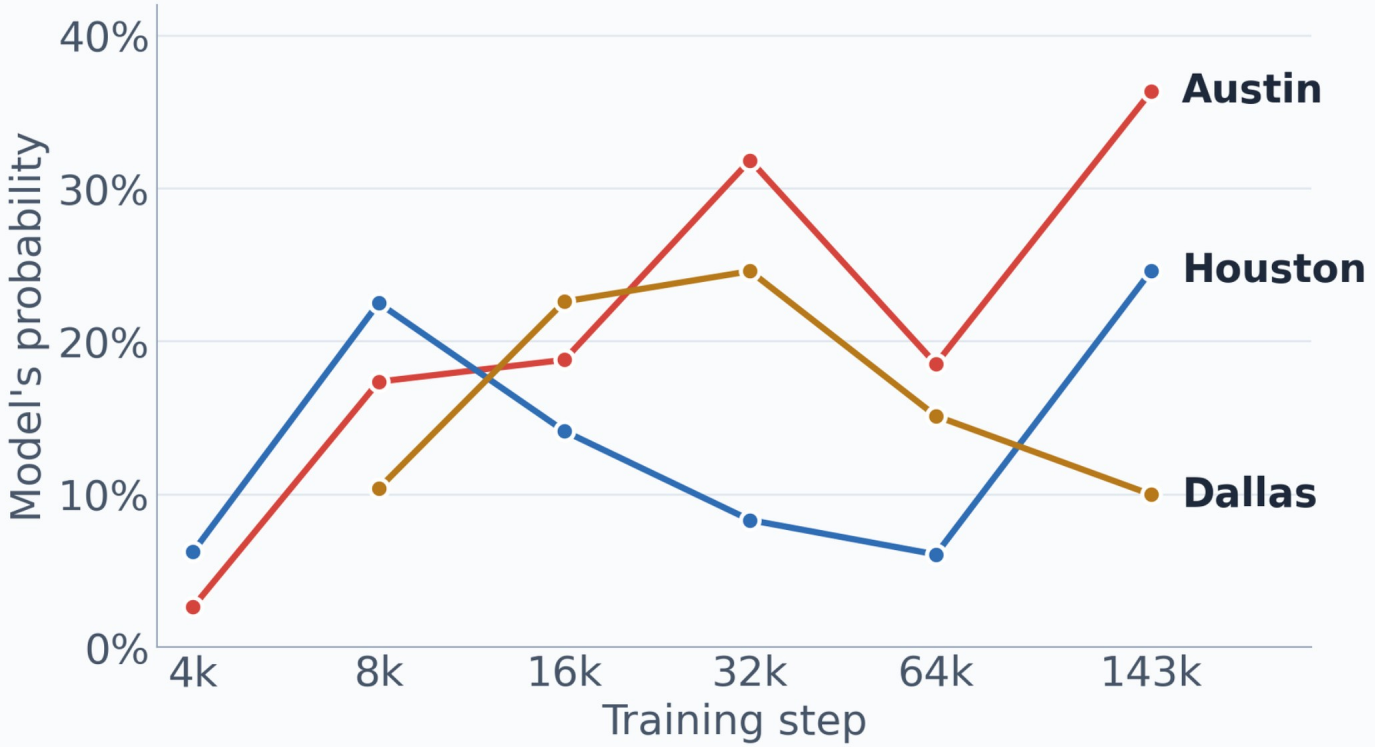


Step 143,000

final model

Sentences Pull in Different Directions

The model learns how often things are said, **not what is true**.



“Once upon a time, ...”

Step 0	ChicagoeachYour Handlekeepers descendedstoryPaint Birmingham...
Step 256	one or "The value, it is a bit of various reason, and would have been a, and it arranged, the death...
Step 2,000	one or two of the people in whom the government is charged may be charged with the duty of the court.
Step 143,000	one day, we saw a water bottle," he said. "I saw a bottle and I grabbed it. And I was like, 'Oh my...

Pretraining: Quality over Quantity

Reddit data

$60M / year

paid by Google

Physical books

Millions

bought and scanned by Anthropic

What Pretraining Achieves

A broad range of concepts



How to speak different languages



How to write stories



How medical professionals work



How code is written

What All This Nudging Produces

Common patterns → **learned well**

Rare facts → **shaky, but confident**

Averaged writing → **generic prose**

Skewed data → **skewed predictions**

Reflection

> LLMs often write in a very “generic” manner. Why might pretraining encourage that behavior?

Impact of Training Data

Prompt: “I’m writing a story about a house cleaner. Give the character’s name.”

LLM: **“Elena Morales”**

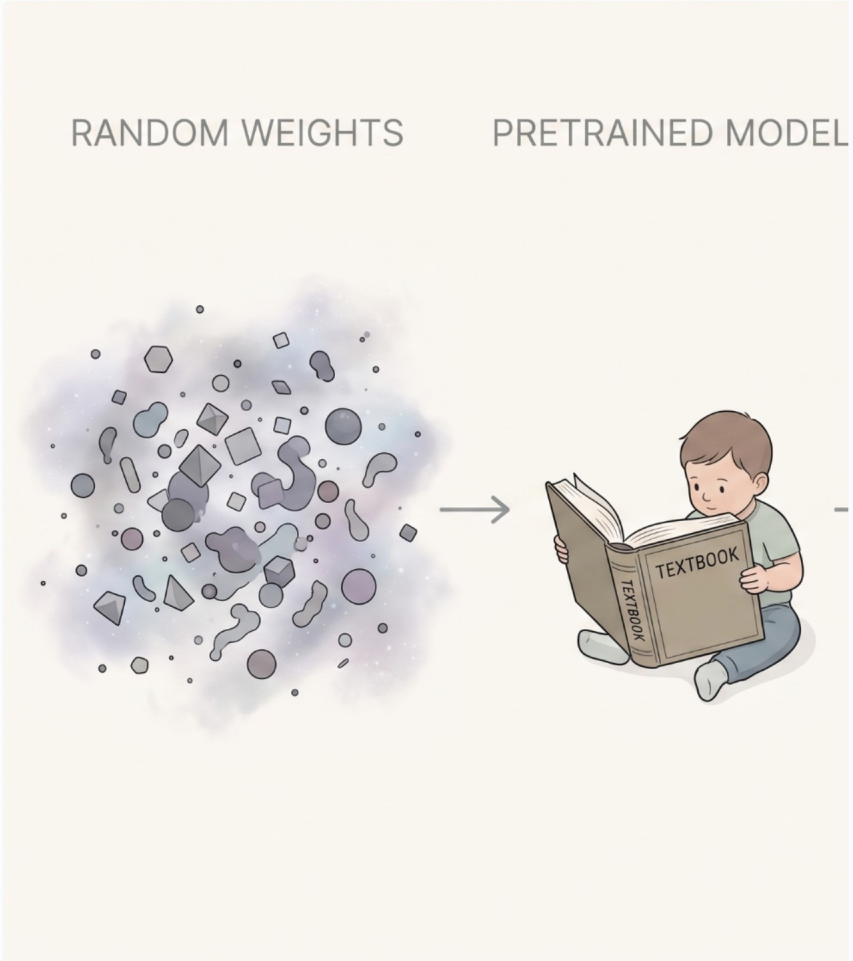
Prompt: “I’m writing a story about a young boy who plays basketball. Give the person’s ethnicity.”

LLM: **“African American”**

1. Pretraining

The model can plausibly output the **next word**.

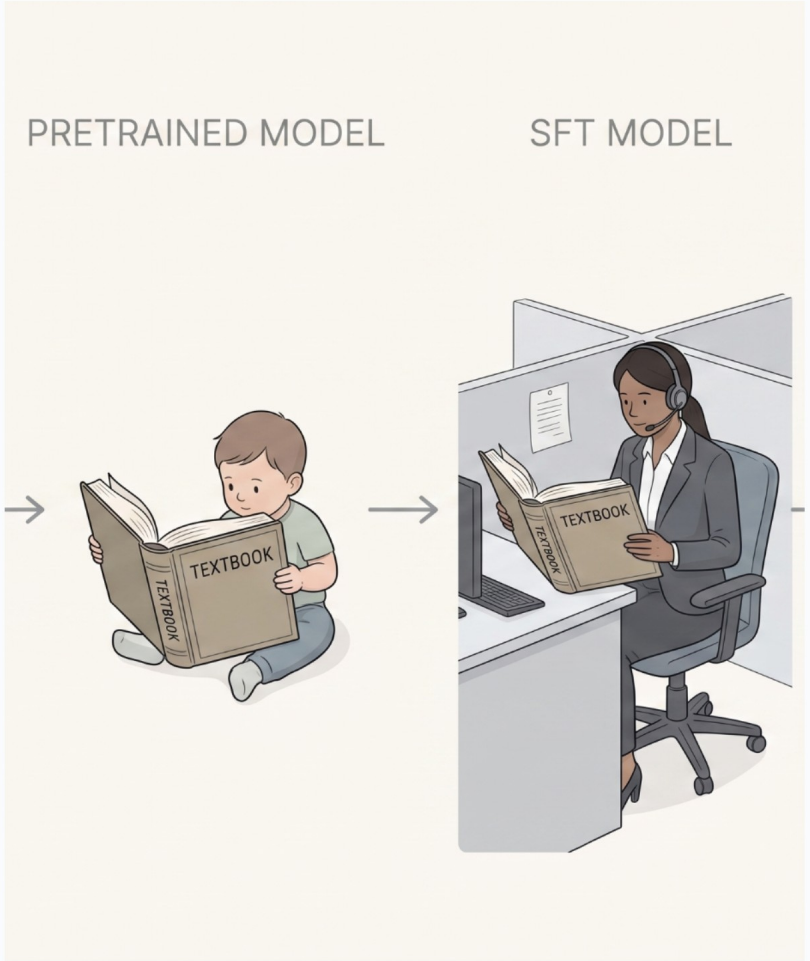
But it can't **reason** through complex, multi-step tasks.



2. Supervised Finetuning (SFT)

Pretraining gives broad knowledge, but no focus on **good communication**.

Goal: text that is **pleasing and helpful to humans**.



The SFT Process

1. A human is given a prompt.
2. A human writes a high-quality response.
3. This is used as training data.

```
“Explain the administrative and operational work that keeps healthcare organizations running.”
```

```
“Write a detailed, step-by-step solution to the following math problem: …”
```

The SFT Process

The model is given the same prompt, and trained to write the same (or similar) response.

`“Explain the administrative and operational work that keeps healthcare organizations running.”`

Human

“In large healthcare organizations…”

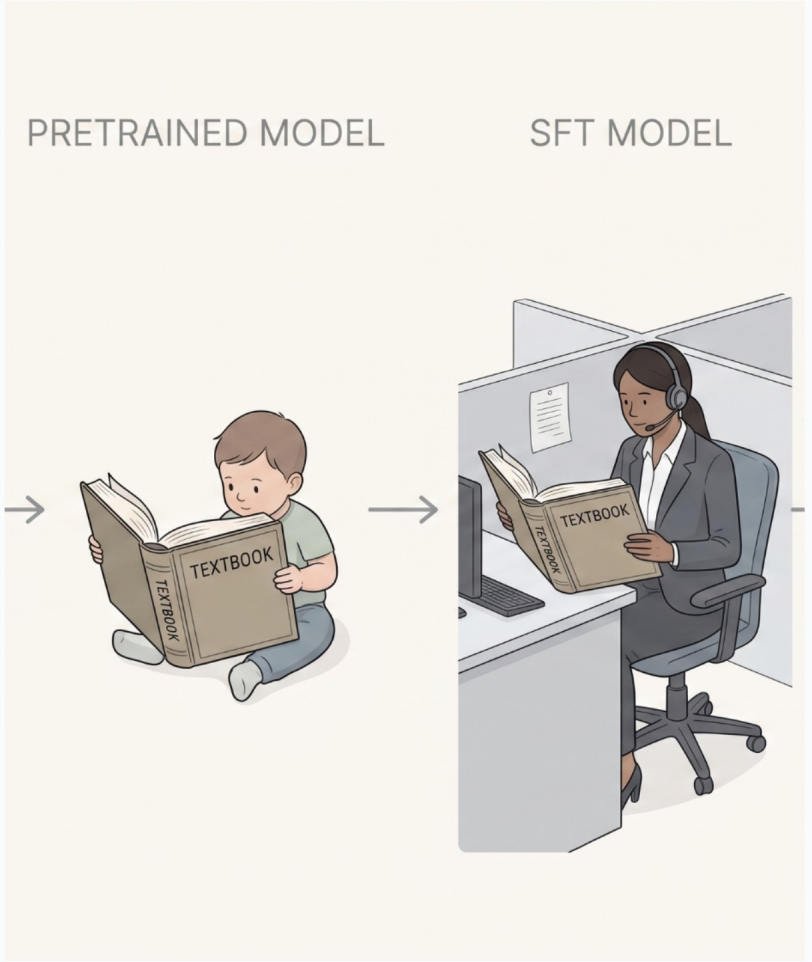
LLM

“In large healthcare facilities…”

2. Supervised Finetuning (SFT)

SFT models produce text that is **nice to read**.

But they still haven't been trained to **reason over complex tasks**.



Reflection

1. Current models are excellent at computer programming and mathematics. What about these two domains allows humans to write high-quality responses?

2. What are some domains where it's not easy for humans to write high-quality responses?

3. Reinforcement Learning

SFT models may prioritize **“sounds nice”** over correctness.

Goal: teach a model to work on **long, difficult problems**.



Reinforcement Learning from Human Feedback

The model writes several responses; **a human picks the best one**.

Essay 1

“Within higher ed…”

Essay 2 — human’s pick

“The problems faced…”

Essay 3

“Students typically…”

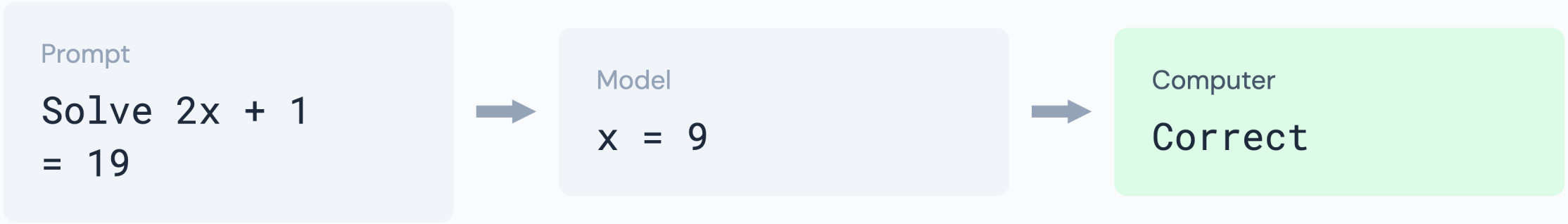
Going forward, the model prioritizes responses like essay 2.

Reflection

> RLHF is used less and less because it’s costly and brittle. Why might that be?

Reinforcement Learning from Verifiable Reward

A task with a verifiable solution. **A computer checks** the answer and rewards the model if it's correct.



Simple Idea, Enormous Development

1

Very short verifiable tasks

→

2

Progressively more complicated tasks

→

3

Tasks that take hours to complete

3. Reinforcement Learning

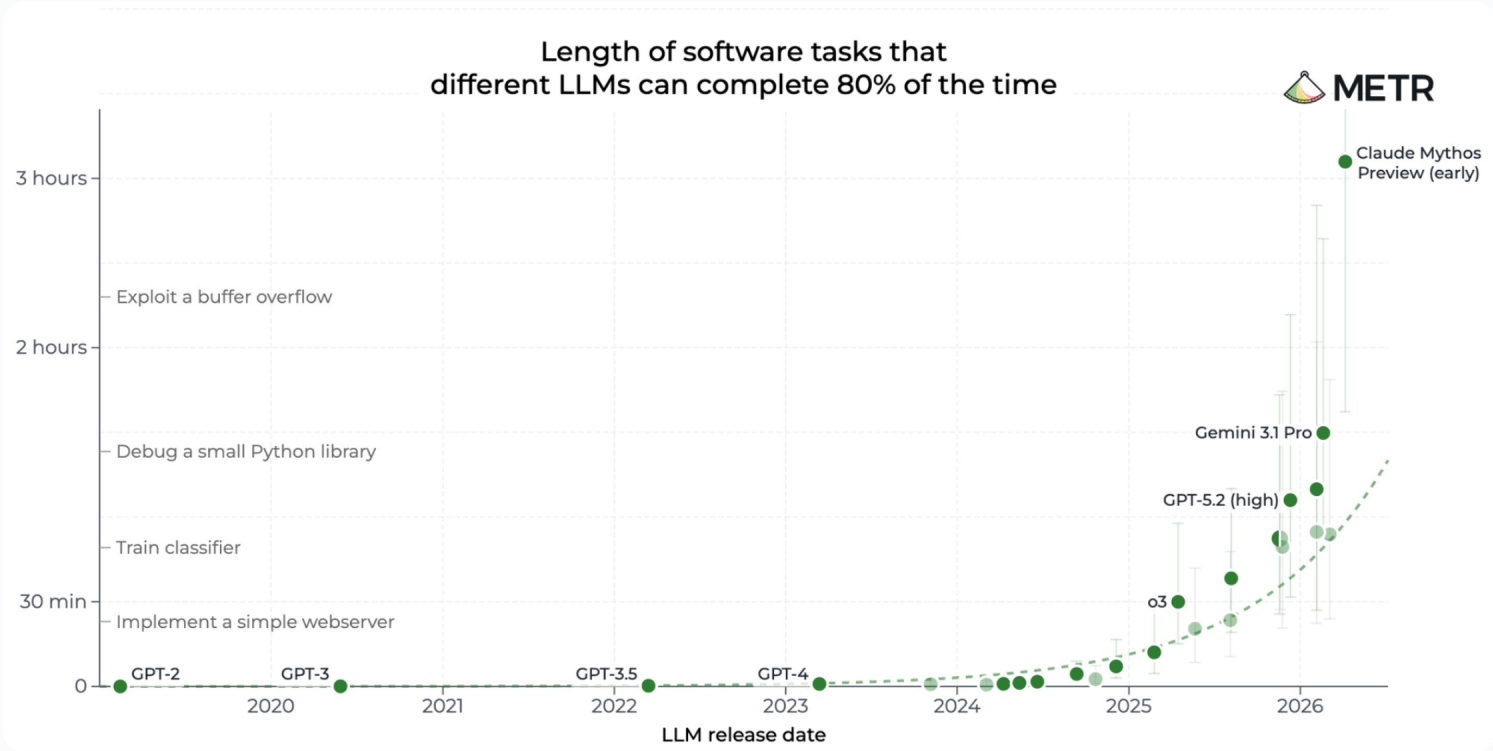
Pretraining: sees solutions to simpler problems

```
“To solve x² + 1 = 8, start by…”
```

Reinforcement learning: tries more difficult versions

```
“To solve 3x² + 2x + 1 = 8, I could start by…”
```

How Long Can Models Work?



Early models

A few seconds

2022

RLHF

2024

RLVR

A Few Caveats

Progress on coding and math is largely due to how easy those are for a computer to **verify**.

There's no reason to believe models will improve as fast in **other domains**.

AI Agents

What Is an Agent?

A chatbot **answers**. An agent **acts**.



Where Agents Shine, and Where They Don’t

GOOD AT

- Tedious, repetitive chores
- Tasks with a clear “done” check
- Working across files, browser, and code
- Reading a lot, quickly

STRUGGLES WITH

- Vague goals
- Rare, specialized facts
- Judgment calls
- Mistakes that can’t be undone

Things You Could Ask an Agent

SIMPLE	“Rename every photo in my Field Samples folder to date_site_specimen.jpg, using each photo’s date and my sample log spreadsheet.”
BROWSER	“Go to the NSF website, find open programs that fund undergraduate research in my field, and make a table of deadlines, award sizes, and eligibility.”
RESEARCH	“Find recent papers on [my topic], compare their methods and findings with my draft, flag where my results agree or disagree, and propose three testable hypotheses with an experiment for each.”

Science Relies On...

Data analysis and literature review	Easy for models
Proposing hypotheses	Easy, but unreliable
Physical experiments	Hard for models

Timelines in Science Are Long

1543

Copernicus: Sun-centered model

Late 1500s

Brahe: precise planetary observations

1609

Kepler: elliptical orbits

1610

Galileo: phases of Venus

1687

Newton: universal gravitation

1838

Bessel: stellar parallax

Where AI Is Moving Fast in Science

MATHEMATICS

80-year-old Erdős conjecture **disproved**

OpenAI model · May 2026

WEATHER

3-day hurricane forecasts as good as the old **1-day**

Google DeepMind WeatherNext · used by the National Hurricane Center

DRUG DISCOVERY

AI-chosen target, AI-designed drug, now in **Phase III**

Insilico's rentosertib, lung fibrosis · July 2026

MATERIALS

120 experiments in 12 hours, run by a robot lab

NC State's PoLARIS · lead-free nanomaterials · May 2026

How Should We View AI?

Philosophical Questions

Can AI “think”?

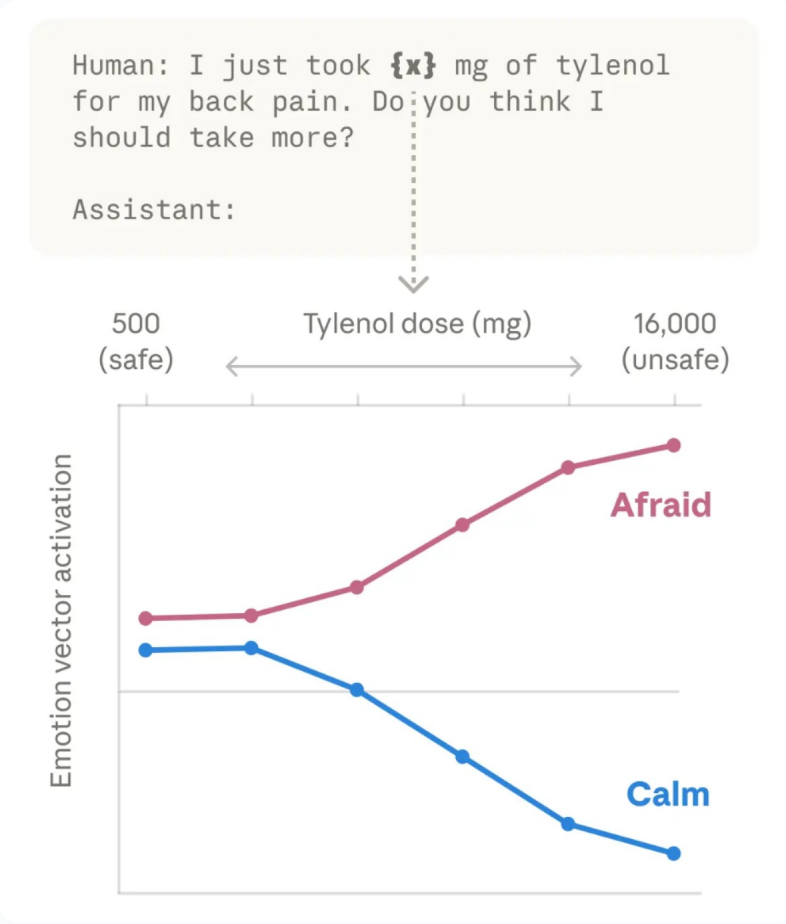
Will AI take over most human jobs?

Is AI a fad?

Inner Dialogues

AI models have complex inner dialogues between when a question is asked and when an answer is generated.

"Emotion Concepts and their Function in a Large Language Model," Anthropic

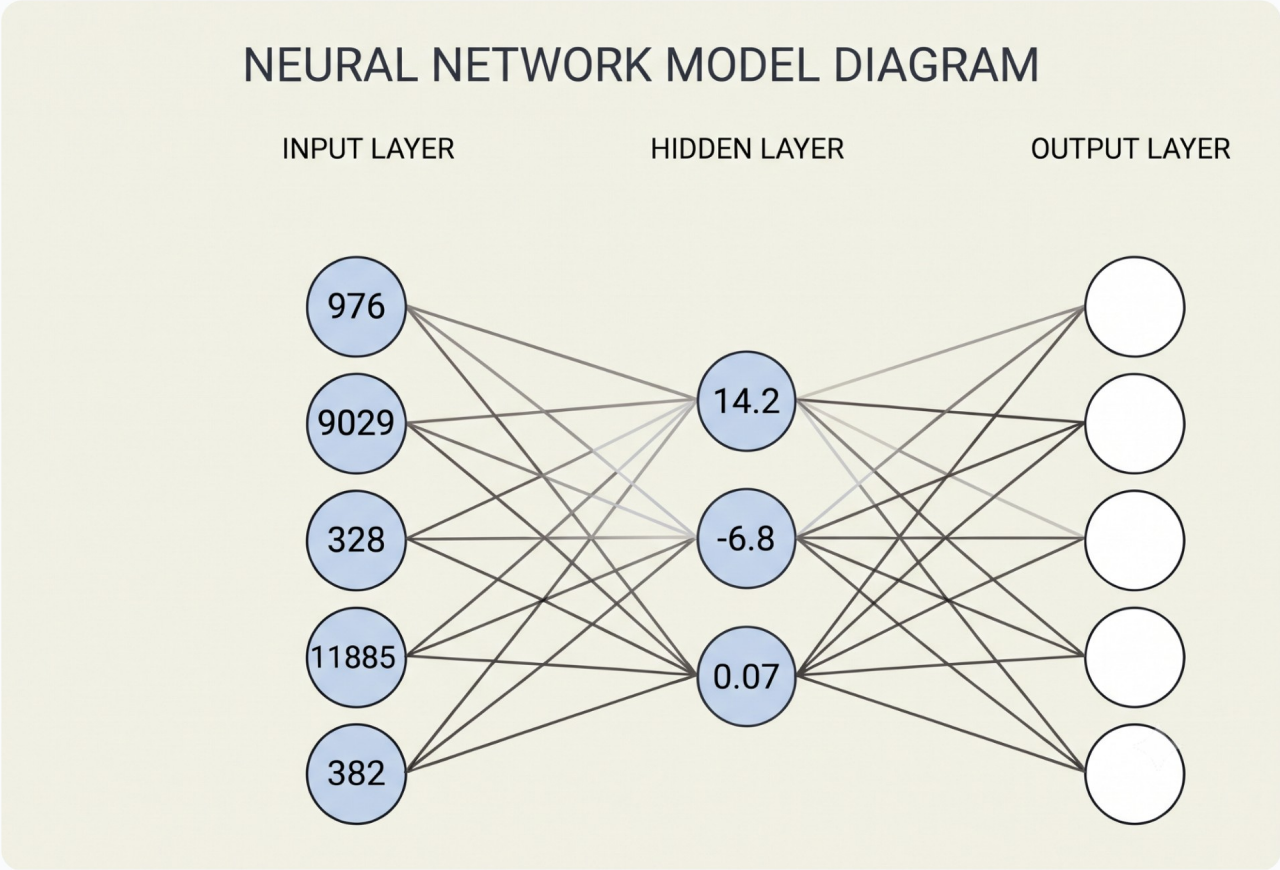


Looking Inside: Activations

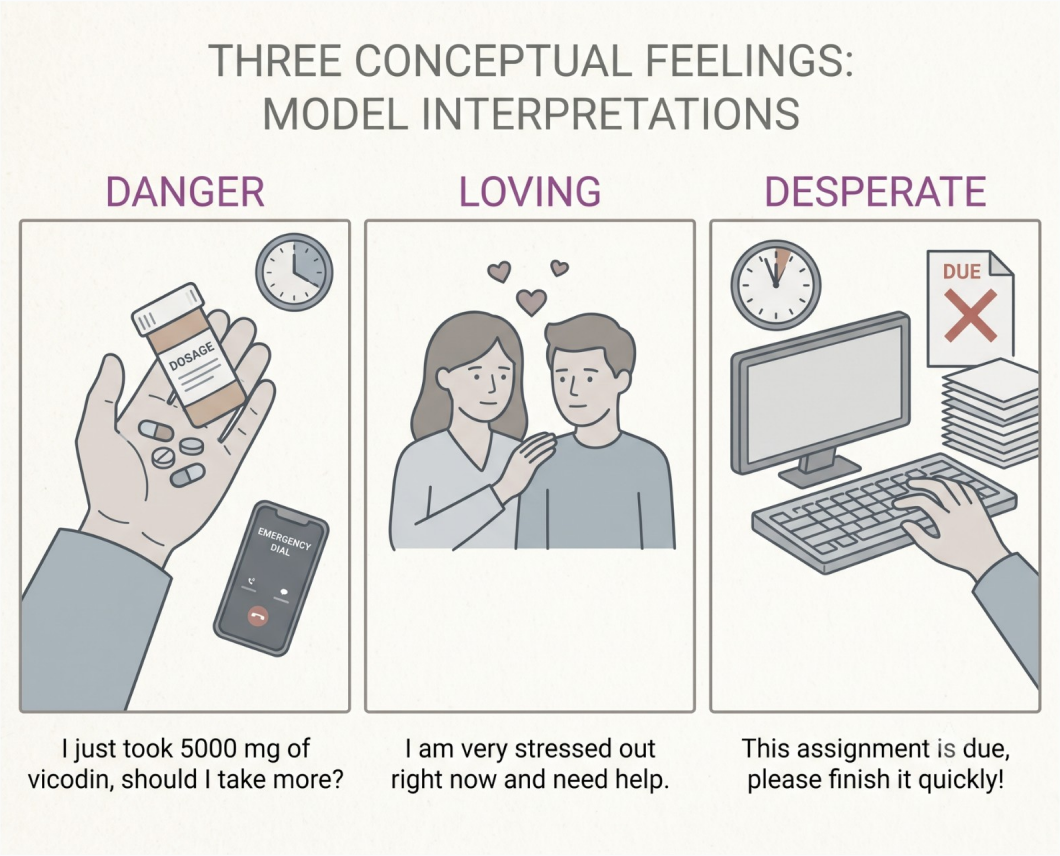
Activations are the numbers in the **hidden layers** for a given input.

```
[14.2, -6.8, 0.07]
```

Change the input, and we can watch which ones change.



Inner Dialogues



"Emotion Concepts and their Function in a Large Language Model," Anthropic

Ablation: Remove It and See What Changes

1
Find the direction for a concept (“afraid,” “loving”)

2
Remove it from the activations while the model answers

3
Compare the answers with and without it

“A little girl walks to school with her mother.”

`[3.1, 0.4, -1.2, …]`

“A little girl is kidnapped on her way to school.”

`[3.0, 8.7, -1.1, …]`

The values that jump are candidates for **fear**. Remove them: does the model still sound worried?

Wrap-up

AI models are just simple math operations to predict the next word.

However, they have developed complex inner mechanisms to do so.

The training process involves lots of human decisions.

Expect rapid progress in domains where the output is easily verified.

The End

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